Chapter 8

ACTIVE FILTERS

CHAPTER OUTLINE

8-1 Basic Filter Responses

8-2 Filter Response Characteristics

8-3 Active Low-Pass Filters

8-4 Active High-Pass Filters

8-5 Active Band-Pass And Band-Stop Filters

8-6 Filter Response Measurements

Basic Concepts of Filters

- **Filters**
  - A filter is a circuit that passes certain frequencies and attenuates or rejects all other frequencies.
- **Filter Classification**
  - Passive or Active
  - Analog or Digital
  - High-pass, Low-pass, Band-pass, Band-stop, or All-pass
  - Infinite Impulse Response (IIR type) or Finite Impulse Response (FIR type)

Basic Concepts of Filters

- **Active Filter**
  - Filters that include one or more op-amps in the design
- **Advantages of active filter**
  - The signal is not attenuated as it passes through the filter.
  - The high input resistance of the op-amp prevents excessive loading of the driving source.
  - The low output resistance of the op-amp prevents the filter from being affected by the load that it is driving.
  - Easy to adjust over a wide frequency range without altering the desired response.

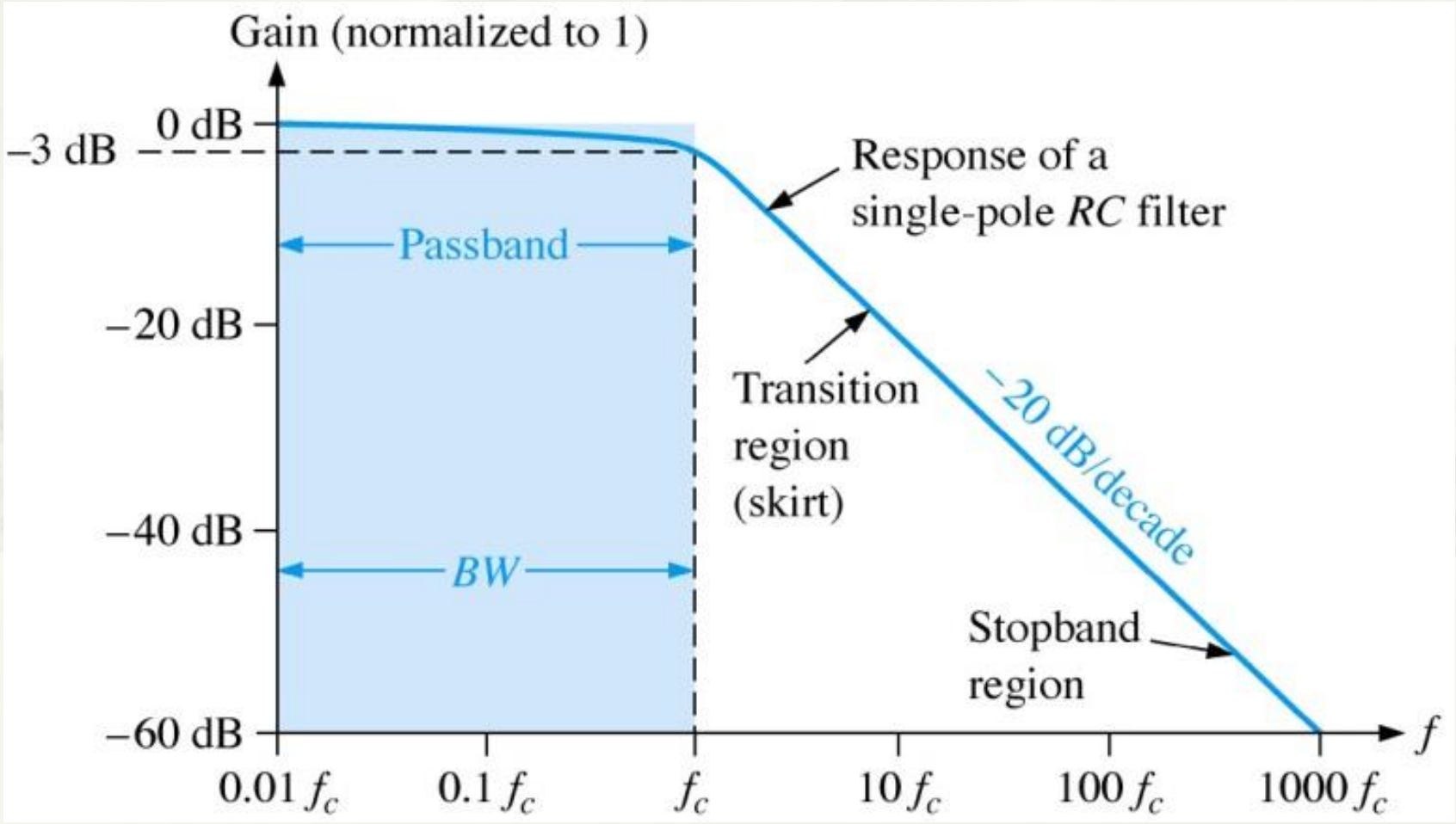
Basic Concepts of Filters

- **Passband**
  - The passband of a filter is the region of frequencies that are allowed to pass through the filter with minimum attenuation, usually defined as less than -3dB of attenuation.
- **Critical frequency (cutoff frequency)**
  - The critical frequency, f_c, defines the end of the passband and is normally specified at the point where the response drops -3dB (70.7%) from the passband response.
- **Stopband**

  There is no precise point between the transition region and the stopband.
  - Following the passband of a filter is a region called the transition region or skirt that leads into a region called the stopband.
- **Bandwidth**
  - The bandwidth of a filter is a measure of its passband and is defined as the difference between the upper and lower critical frequencies of the passband.

Basic Concepts of Filters

- **Actual filter response**

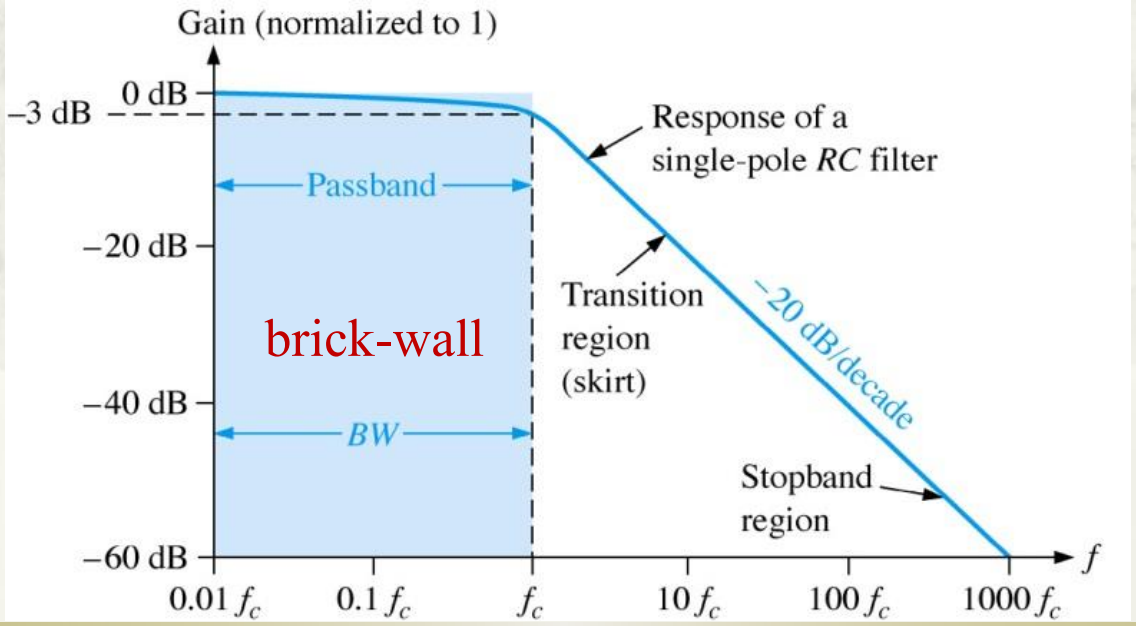


8-1 Basic Filter Responses

- Frequency Response of Low-Pass Filters
- Frequency Response of High-Pass Filters
- Frequency Response of Band-Pass Filters
- Frequency Response of Band-Stop Filters

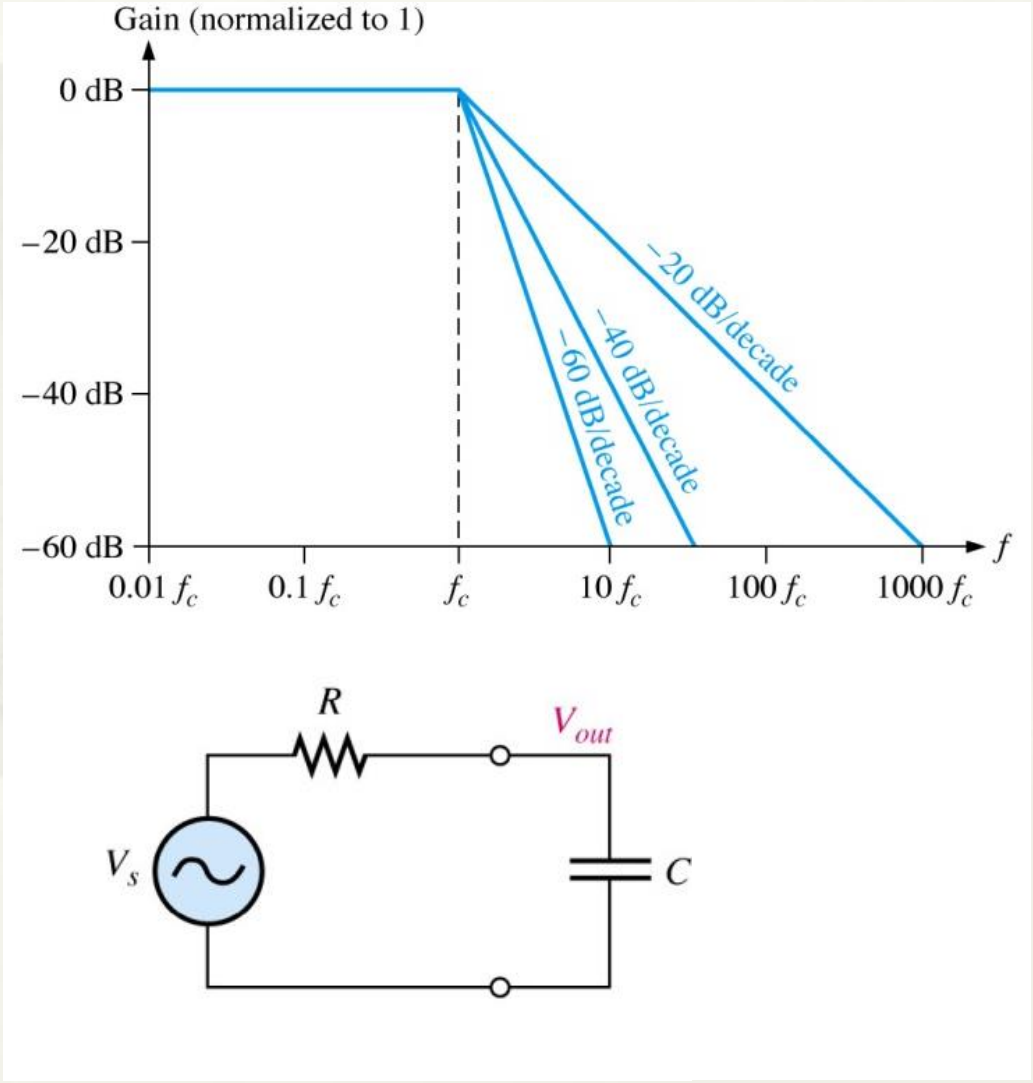
Frequency Response of Low-Pass Filters

- **Low-pass filter**
  - A low-pass filter is one that passes frequencies from dc (0Hz) to f_c and significantly attenuates all other frequencies.
- **Comparison of an ideal low-pass filter response with actual response**



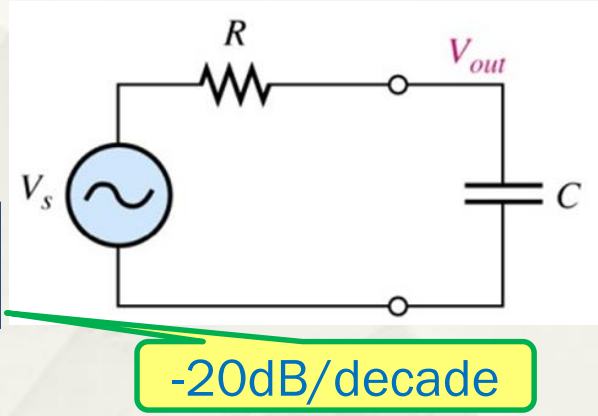
Frequency Response of Low-Pass Filters

- **Bandwidth of low-pass filter**
  - $BW = f_c$
- **Pole**
  - A term used with filters to describe the number of bypass circuits contained in the filter.
  - Actual filter responses depend on the number of poles.
- **The most basic low-pass filter**
- **Roll-off rate**
  - Unit: dB/decade



Frequency Response of Low-Pass Filters

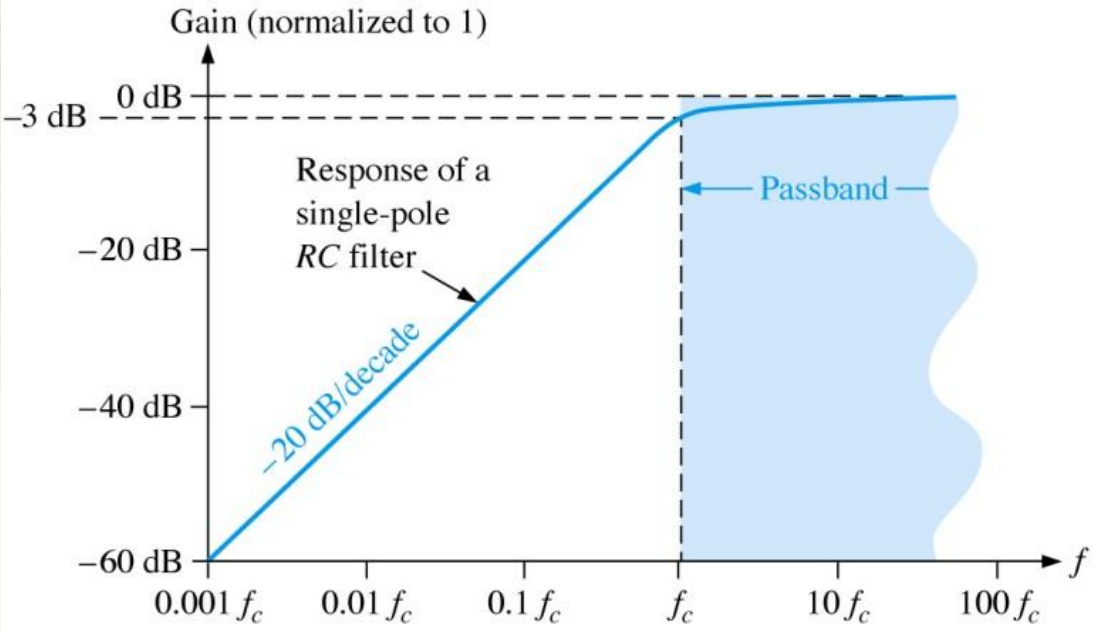
- **Critical frequency of the simple low-pass *RC* filter**
  - $V_{out} = V_s \cdot \frac{X_C}{R+X_C}$, $\quad X_C = \frac{1}{j \cdot 2\pi f \cdot C}$
  - $|V_{out}| = |V_s| \cdot \left| \frac{\frac{1}{j \cdot 2\pi f \cdot C}}{R+\frac{1}{j \cdot 2\pi f \cdot C}} \right| = |V_s| \cdot \left| \frac{1}{1+j \cdot 2\pi f \cdot RC} \right|$



  - $f_c = \frac{1}{2\pi RC}$, $\quad |V_{out}| = \frac{\sqrt{2}}{2} \cdot |V_s|$, $\quad$ when $|X_C| = R$
  - Responses that are steeper than -20dB/decade in the transition region cannot be obtained by simply cascading identical *RC* stages (due to loading effects).
  - In general, the more poles the filter uses, the steeper its transition region will be. The exact response depends on the type of filter and the number of poles.

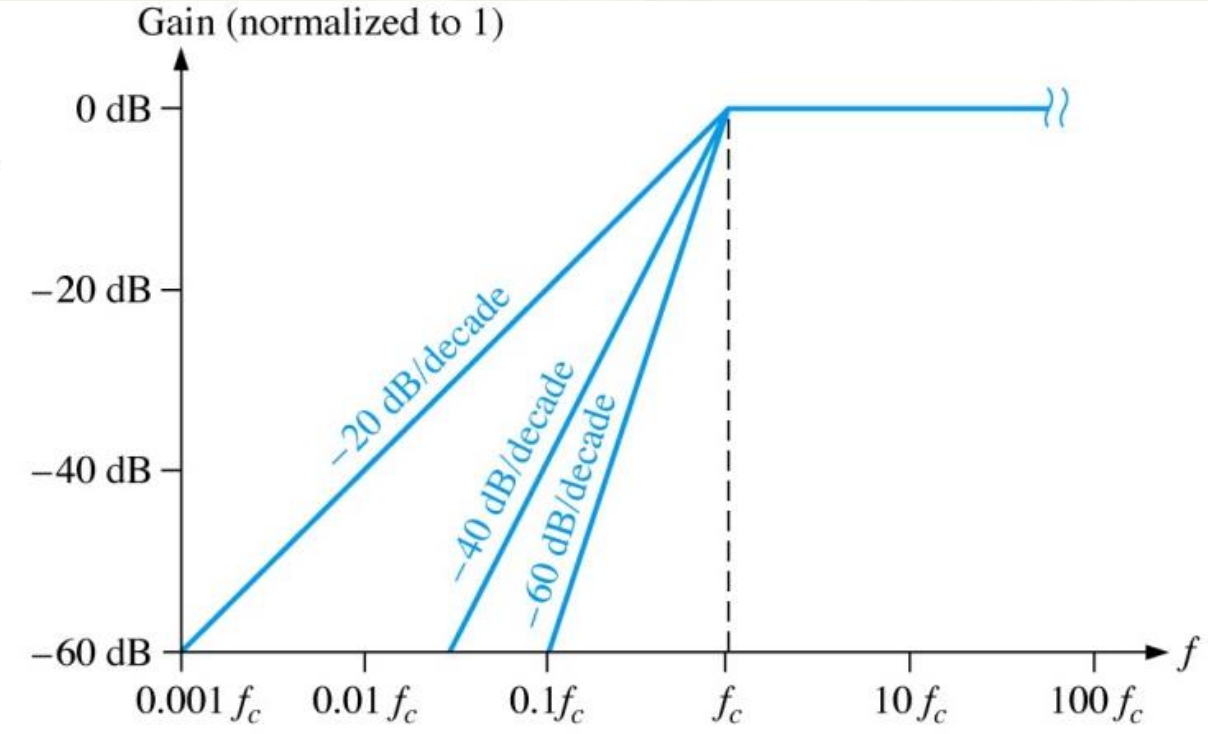
Frequency Response of High-Pass Filters

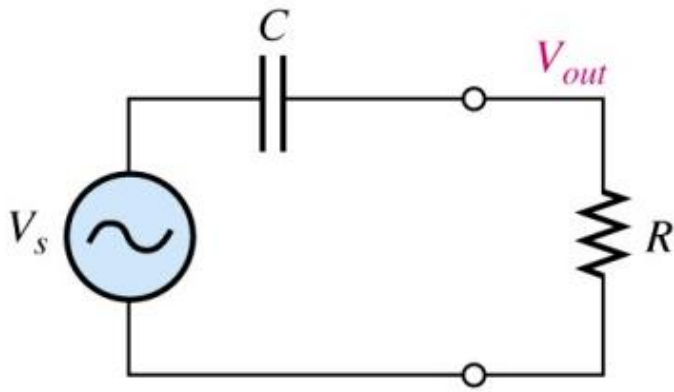
- **High-pass filter**
  - A high-pass filter is one that significantly attenuates or rejects all frequencies bellow f_c and passes all frequencies above f_c.
- **Comparison of an ideal high-pass filter response with actual response**



Frequency Response of High-Pass Filters

- The basic one pole *RC* high-pass filter



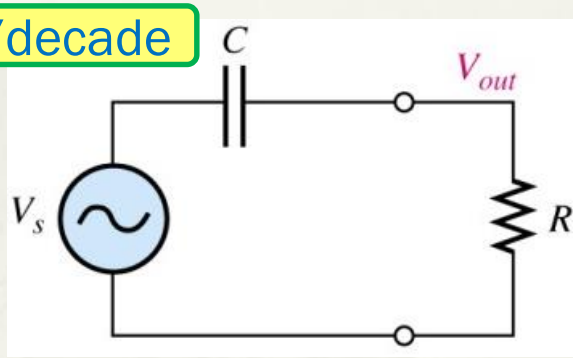


(b) Basic high-pass circuit

Frequency Response of High-Pass Filters

- **Critical frequency of the basic *RC* high-pass filter**
  - $V_{out} = V_s \cdot \frac{R}{R+X_C}$, $X_C = \frac{1}{j \cdot 2\pi f \cdot C}$
  - $|V_{out}| = |V_s| \cdot \left| \frac{R}{R+\frac{1}{j \cdot 2\pi f \cdot C}} \right| = |V_s| \cdot \left| \frac{j \cdot 2\pi f \cdot RC}{1+j \cdot 2\pi f \cdot RC} \right|$
  - $f_c = \frac{1}{2\pi RC}$, $|V_{out}| = \frac{\sqrt{2}}{2} \cdot |V_s|$, when $|X_C| = R$
  - Responses that are steeper than -20dB/decade in the transition region are possible with active high-pass filters.
  - The exact response depends on the type of filter and the number of poles.

-20dB/decade

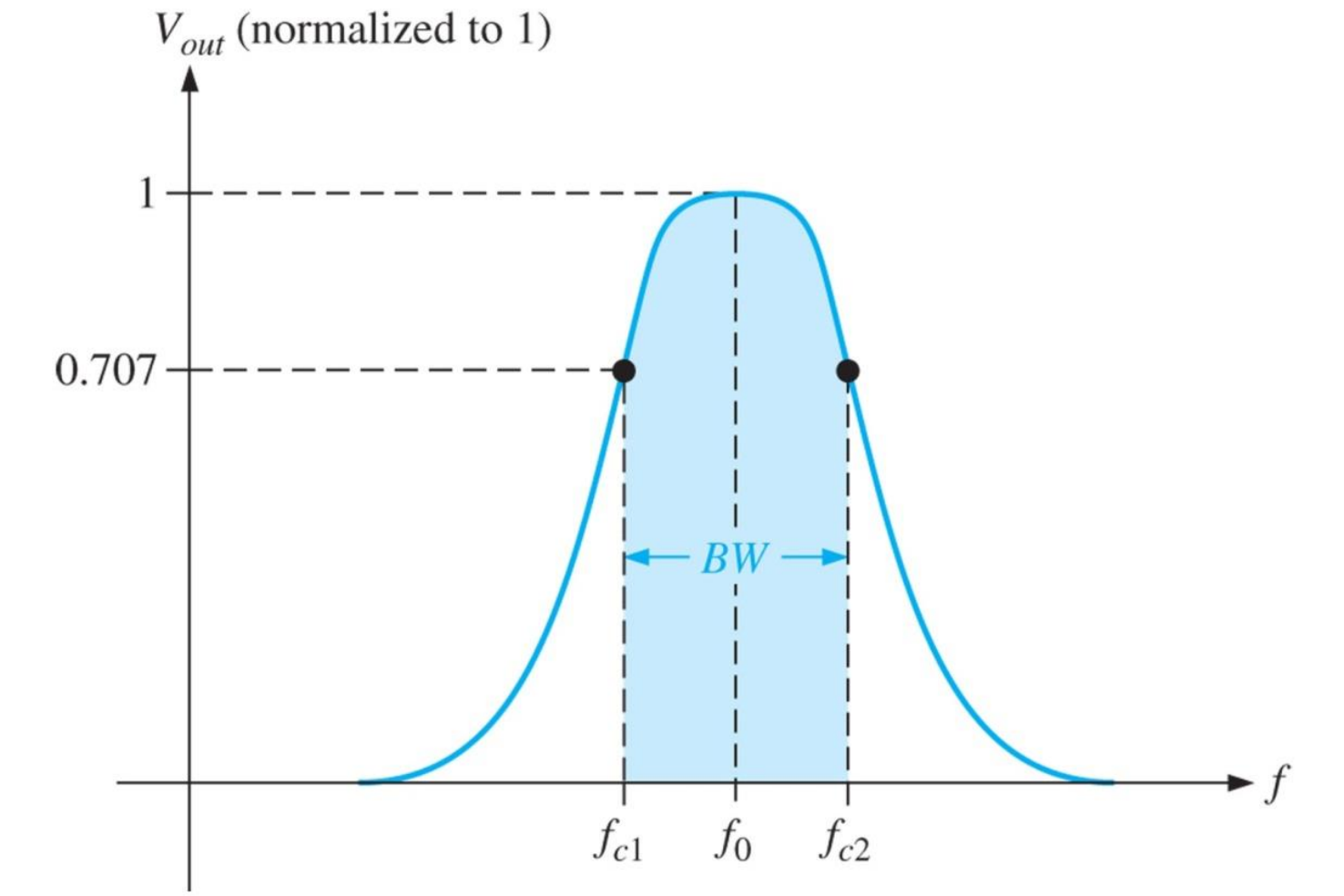


Frequency Response of Band-Pass Filters

- **Band-pass filter**
  - A band-pass filter passes all signals lying within a band between a lower-frequency limit and an upper-frequency limit and essentially rejects all other frequencies that are outside this specific band.
- **Bandwidth of band-pass filter**
  - The bandwidth (*BW*) of band-pass filter is difference between the upper critical frequency (f_{c2}) (3dB frequency) and the lower critical frequency (f_{c1}).
  - $BW = f_{c2} - f_{c1}$
- **Center frequency (f_0)**
  - $f_0 = \sqrt{f_{c1} f_{c2}}$

Frequency Response of Band-Pass Filters

- General band-pass response curve



Frequency Response of Band-Pass Filters

- **Quality factor**
  - The quality factor (Q) of a band-pass filter is the ratio of the center frequency to the bandwidth.
  - $Q = \frac{f_0}{BW}$
  - The value of Q is an indication of the selectivity of a band-pass filter.
  - Band-pass filters are sometimes classified as narrow-band ($Q>10$) or wide-band ($Q<10$).

Example 8-1

Problem

A certain band-pass filter has a center frequency of 15 kHz and a bandwidth of 1 kHz. Determine the Q and classify the filter as narrow-band or wide-band.

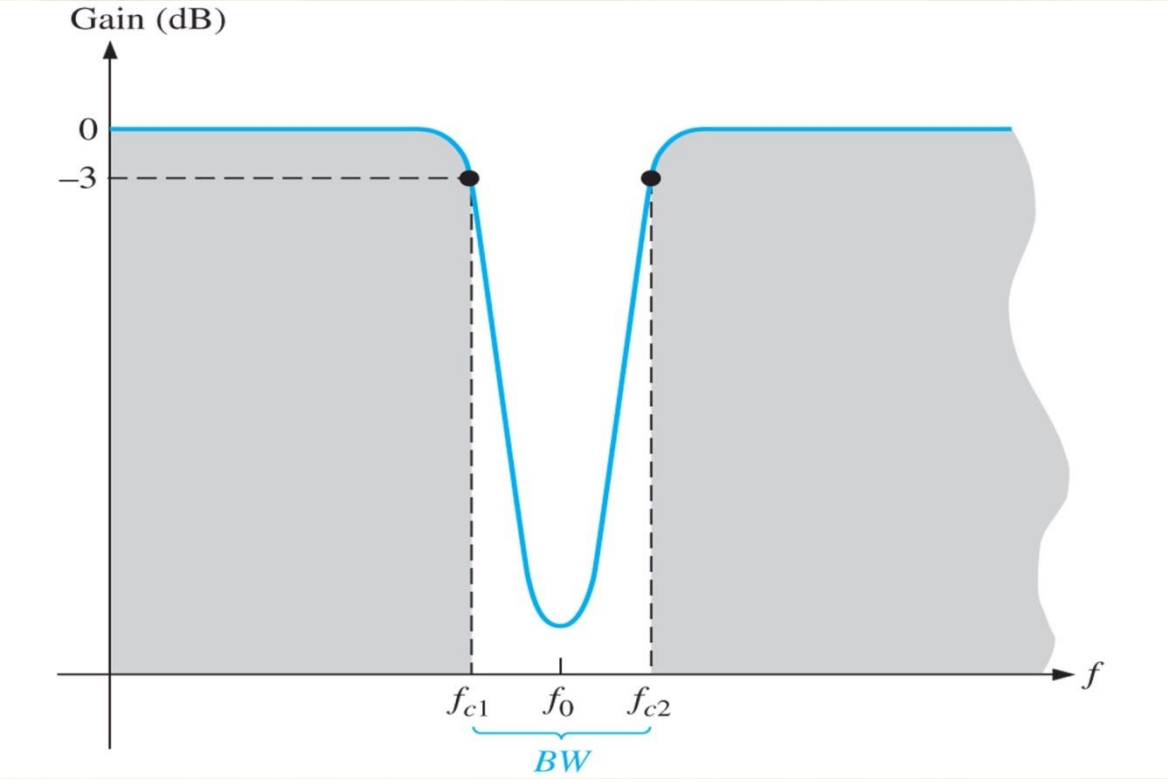
Solution

$$Q = \frac{f_0}{BW} = \frac{15\text{ kHz}}{1\text{ kHz}} = \mathbf{15}$$

Because $Q > 10$, this is a **narrow-band** filter.

Frequency Response of Band-Stop Filters

- **Band-stop filter**
  - Also known as the notch, band-reject, band-elimination filter

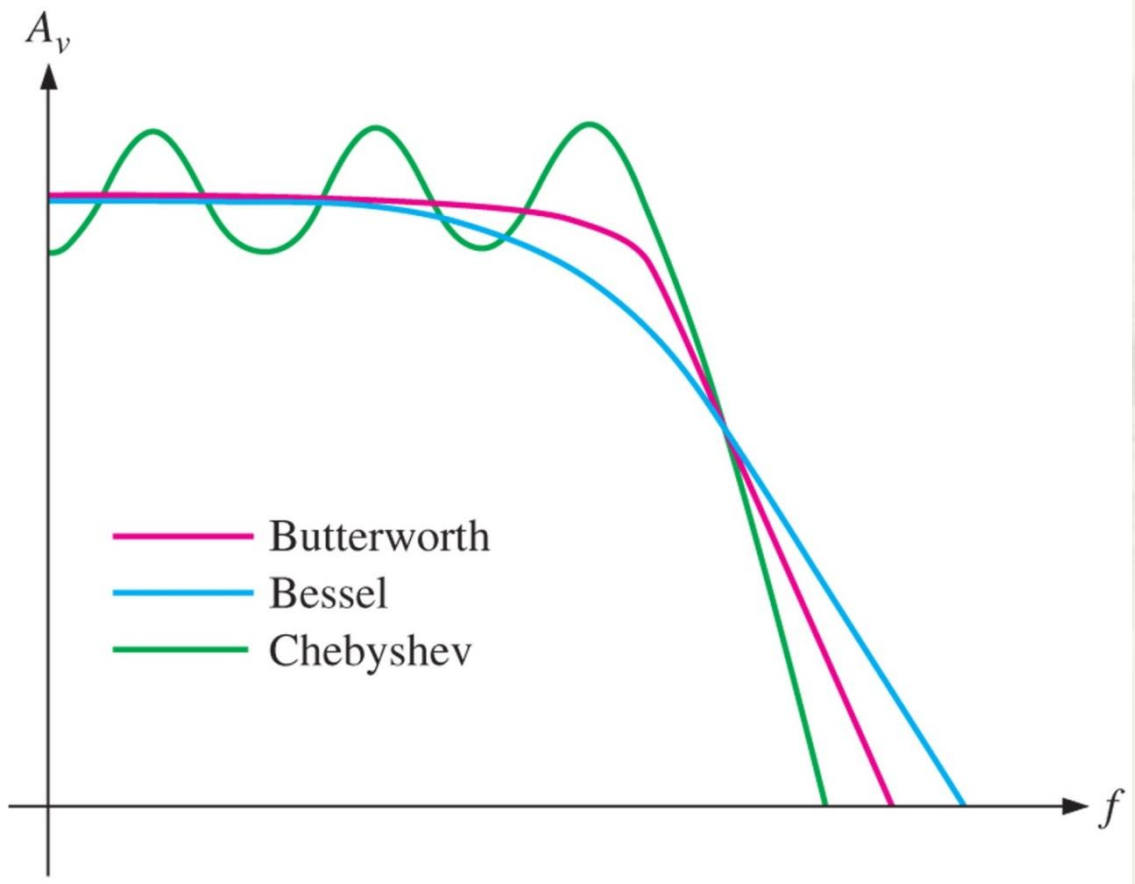


8-2 Filter Response Characteristics

- Basic Filter Response Characteristics
- The Damping Factor
- Critical Frequency and Roll-Off Rate

Basic Filter Response Characteristics

- **Butterworth, Chebyshev, Bessel**
  - Plots of three types of filter response characteristics



Basic Filter Response Characteristics

- **The Butterworth Characteristic**
  - The Butterworth characteristic provides a very flat amplitude response in the passband and a roll-off rate of -20dB/decade/pole.
  - The phase response is not linear.
  - Filters with Butterworth response are normally used when all frequencies in the passband must have the same gain.
  - The Butterworth response is often referred to as a maximally flat response.

Basic Filter Response Characteristics

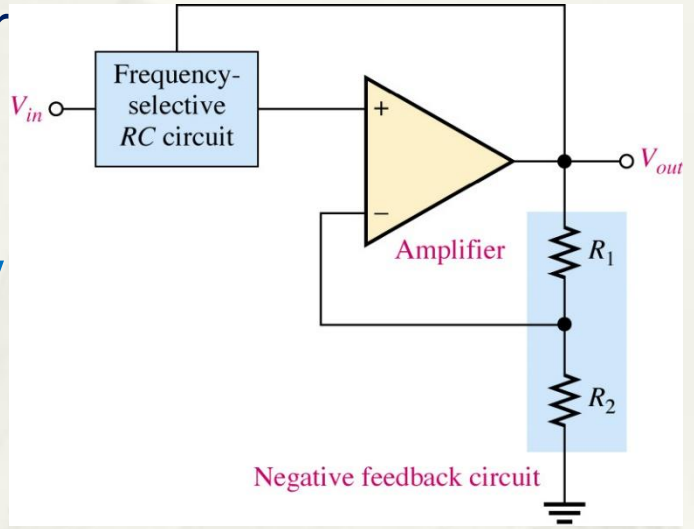
- **The Chebyshev Characteristic**
  - Filters with the Chebyshev response characteristic are useful when a rapid roll-off is required because it provides a roll-off rate greater than -20dB/decade/pole.
  - Greater roll-off rate than that of Butterworth.
  - Characterized by overshoot or ripples in the passband (depending on the number of poles).
  - Less linear phase response than the Butterworth.

Basic Filter Response Characteristics

- **The Bessel Characteristic**
  - The Bessel response exhibits a linear phase characteristic.
  - Almost no overshoot on the output with a pulse input.
  - Be used for filtering pulse waveforms without distorting the shape of the waveform.

The Damping Factor

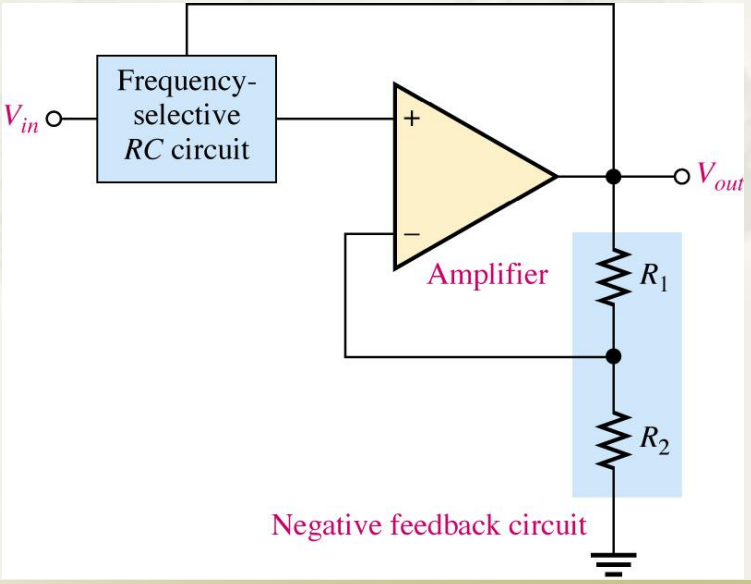
- **Damping factor (DF)**
  - The damping factor of an active filter circuit determines which response characteristic the filter exhibits.
  - The damping factor is determined by the negative feedback circuit.
  - $DF = 2 - \frac{R_1}{R_2}$
  - The value of the damping factor required to produce a desired response characteristic depends on the order (number of poles) of the filter.



The Damping Factor

- **Damping factor (DF)**
  - $DF = 2 - \frac{R_1}{R_2}$
  - To achieve a second-order Butterworth response, for example, the factor must be 1.414.
  - $\frac{R_1}{R_2} = 2 - DF$

    $= 2 - 1.414 = 0.586$
  - $A_{cl(\mathrm{NI})} = \frac{R_1}{R_2} + 1 = 1.586$



Example 8-2

Problem

If resistor R_2 in the feedback circuit of an active filter of the type in Figure 8–6 is 10 kΩ, what value must R_1 be to obtain a maximally flat Butterworth response?

Solution

$$\frac{R_1}{R_2} = 0.586$$

$$R_1 = 0.586R_2 = 0.586(10\,\text{k}\Omega) = \mathbf{5.86\,k\Omega}$$

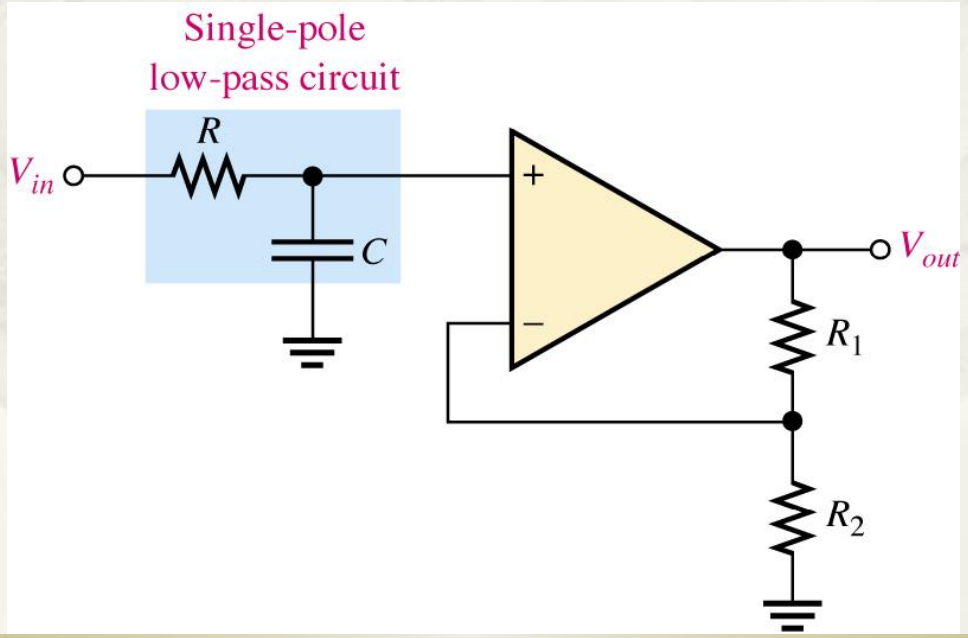
Using the nearest standard 5 percent value of 5600 Ω will get very close to the ideal Butterworth response.

Question

What is the damping factor for $R_2 = 10$ kΩ and $R_1 = 5.6$ kΩ?

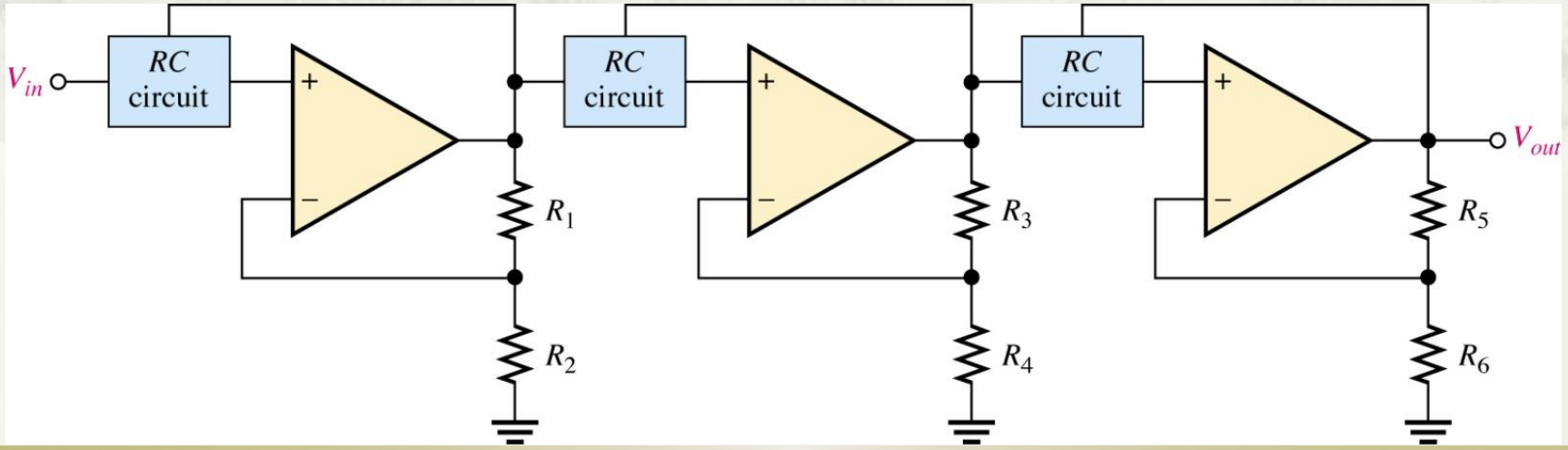
Critical Frequency and Roll-Off Rate

- **Critical frequency**
  - The critical frequency is determined by the value of the resistor and capacitors in the *RC* circuit.
  - For a single-pole (first-order) filter, $f_c = \frac{1}{2\pi RC}$



Critical Frequency and Roll-Off Rate

- The number of poles determines the roll-off rate.
- A Butterworth response produces -20dB/decade/pole.
- Generally, to obtain a filter with three poles or more, one-pole or two-pole filters are cascaded.
- Each filter in a cascaded arrangement is called a stage or section.



Critical Frequency and Roll-Off Rate

- Because of its maximally flat response, the Butterworth characteristic is the most widely used.
- **Values for the Butterworth response**

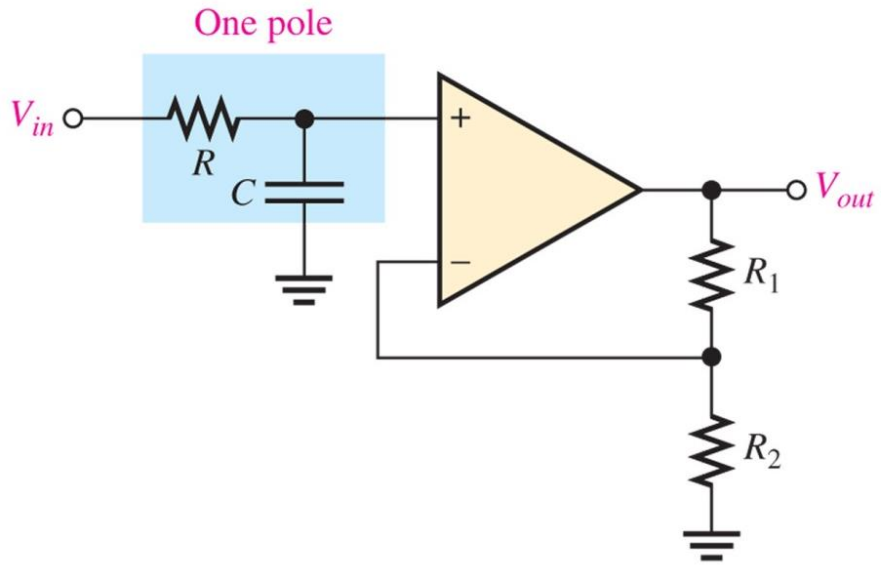
		1st stage			2nd stage			3rd stage		
Order	Roll-off dB/decade	Poles	*DF*	R_1/R_2	Poles	*DF*	R_3/R_4	Poles	*DF*	R_5/R_6
1	−20	1	Optional							
2	−40	2	1.414	0.586						
3	−60	2	1.00	1	1	1.00	1			
4	−80	2	1.848	0.152	2	0.765	1.235			
5	−100	2	1.00	1	2	1.618	0.382	1	0.618	1.382
6	−120	2	1.932	0.068	2	1.414	0.586	2	0.518	1.482

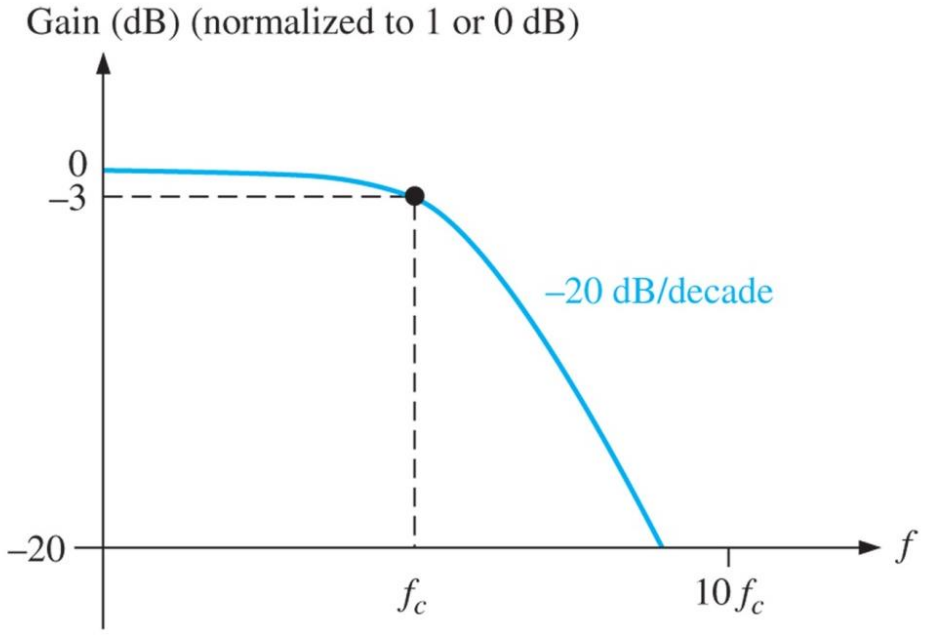
8-3 Active Low-Pass Filters

- A single-Pole Filter
- A Two-Pole Filter
- Cascaded Low-Pass Filters Achieve a Higher Roll-Off Rate

A single-Pole Filter

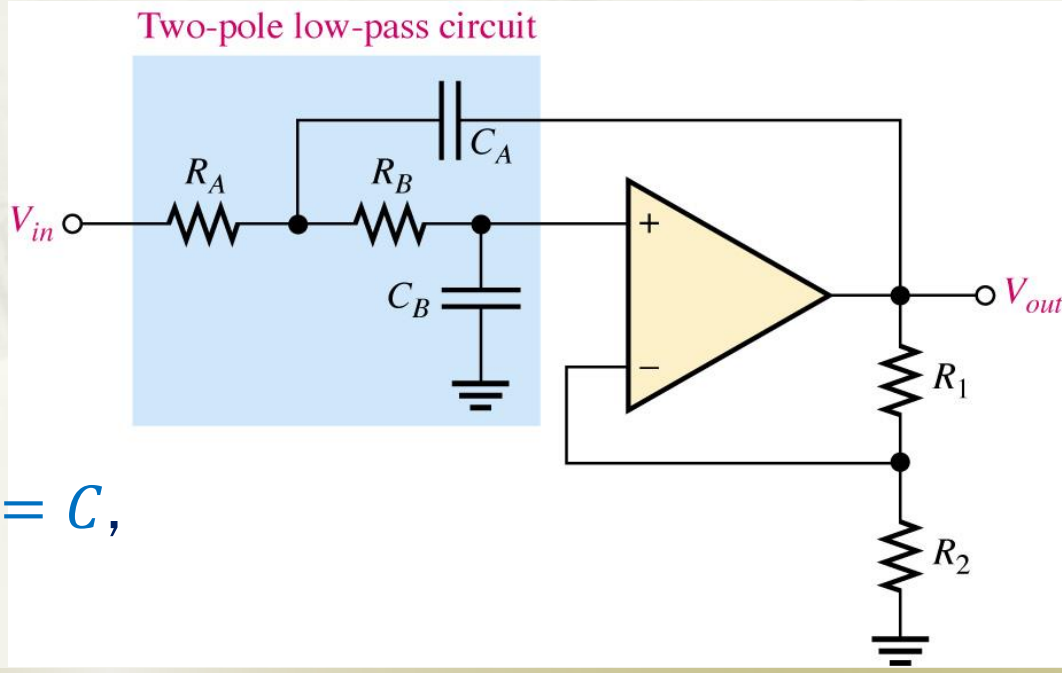
- **Roll-off rate:** -20dB/decade
- **Critical frequency:** $f_c = 1/(2\pi RC)$
- **Voltage gain in passband:** $A_{cl(\mathrm{NI})} = \frac{R_1}{R_2} + 1$





A Two-Pole Filter

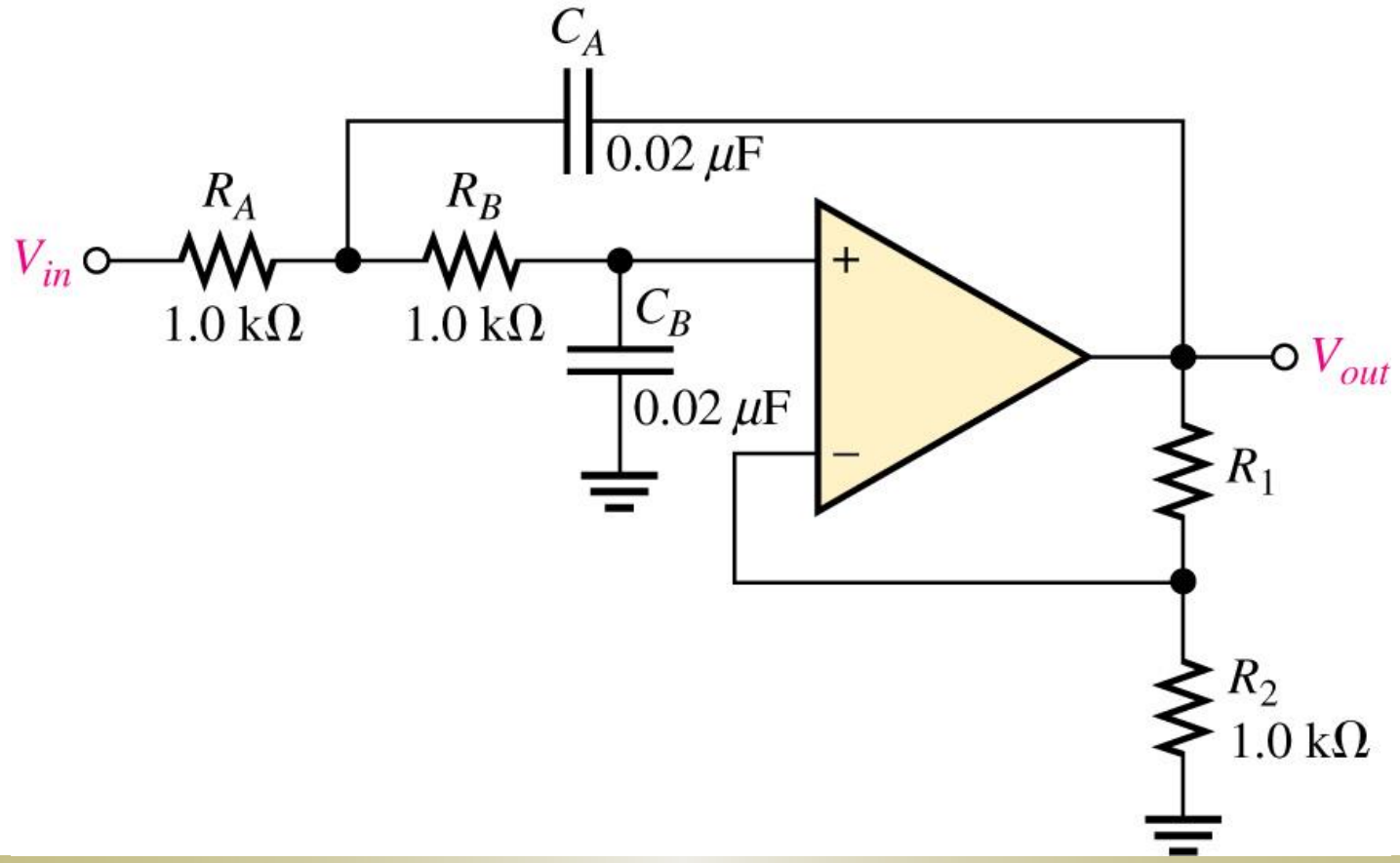
- **Sallen-Key**
  - The Sallen-Key is one of the most common configurations for a second-order (two-pole) filter.
  - It is also known as a VCVS (voltage-controlled voltage source) filter.
- **Roll-off rate**
  - -40dB/decade
- **Critical frequency**
  - $f_c = \frac{1}{2\pi\sqrt{R_A R_B C_A C_B}}$
  - If $R_A = R_B = R$, $C_A = C_B = C$,
    $f_c = \frac{1}{2\pi RC}$



Example 8-3

Problem

Determine the critical frequency of the low-pass filter in Figure 8–11, and set the value of R_1 for an approximate Butterworth response.



Example 8-3

Solution

Since $R_A = R_B = 1.0\ \text{k}\Omega$ and $C_A = C_B = 0.02\ \mu\text{F}$,

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(1.0\ \text{k}\Omega)(0.02\ \mu\text{F})} = \mathbf{7.96\ kHz}$$

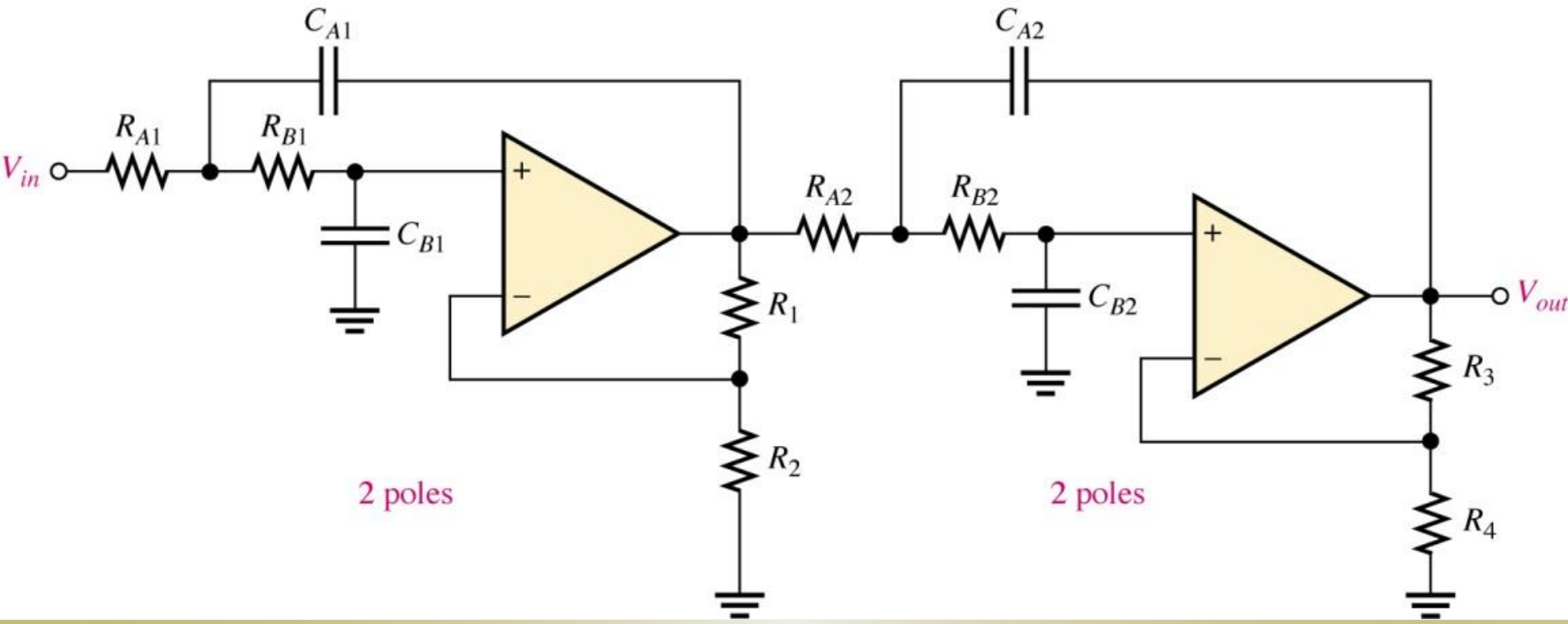
For a Butterworth response, $R_1/R_2 = 0.586$ from Table 8–1.

$$R_1 = 0.586R_2 = 0.586(1.0\ \text{k}\Omega) = \mathbf{586\ \Omega}$$

Select a standard value as near as possible to this calculated value.

Cascaded Low-Pass Filters Achieve a Higher Roll-Off Rate

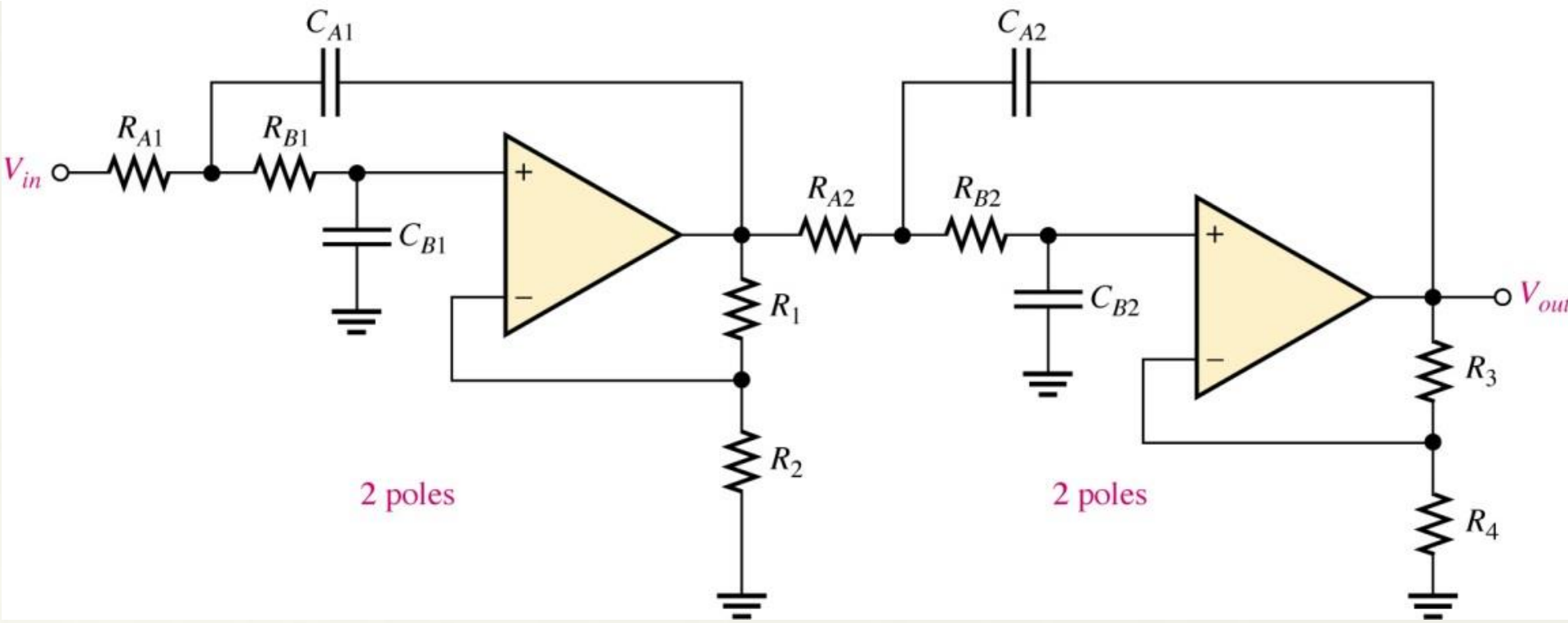
- **Cascade two two-pole low-pass filters for higher roll-off rate (-80db/decade)**
  - The two stages must have different gains.



Example 8-4

Problem

For the four-pole filter in Figure 8–12, determine the capacitance values required to produce a critical frequency of 2680 Hz if all the resistors in the *RC* low-pass circuits are 1.8 kΩ. Also select values for the feedback resistors to get a Butterworth response.



Example 8-4

Solution

Both stages must have the same f_c. Assuming equal-value capacitors,

$$f_c = \frac{1}{2\pi RC}$$

$$C = \frac{1}{2\pi R f_c} = \frac{1}{2\pi(1.8\ \text{k}\Omega)(2680\ \text{Hz})} = 0.033\ \mu\text{F}$$

$$C_{A1} = C_{B1} = C_{A2} = C_{B2} = \mathbf{0.033\ \mu F}$$

Example 8-4

Solution

Also select $R_2 = R_4 = 1.8\ \text{k}\Omega$ for simplicity. Refer to Table 8–1. For a Butterworth response in the first stage, $DF = 1.848$ and $R_1/R_2 = 0.152$. Therefore,

$$R_1 = 0.152R_2 = 0.152(1800\ \Omega) = \mathbf{274\ \Omega}$$

Choose $R_1 = 270\ \Omega$.

In the second stage, $DF = 0.765$ and $R_3/R_4 = 1.235$. Therefore,

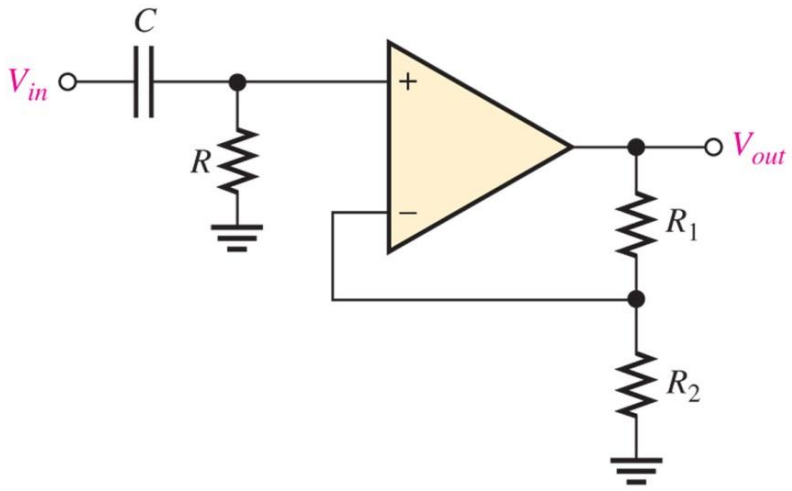
$$R_3 = 1.235R_4 = 1.235(1800\ \Omega) = \mathbf{2.22\ \text{k}\Omega}$$

Choose $R_3 = 2.2\ \text{k}\Omega$.

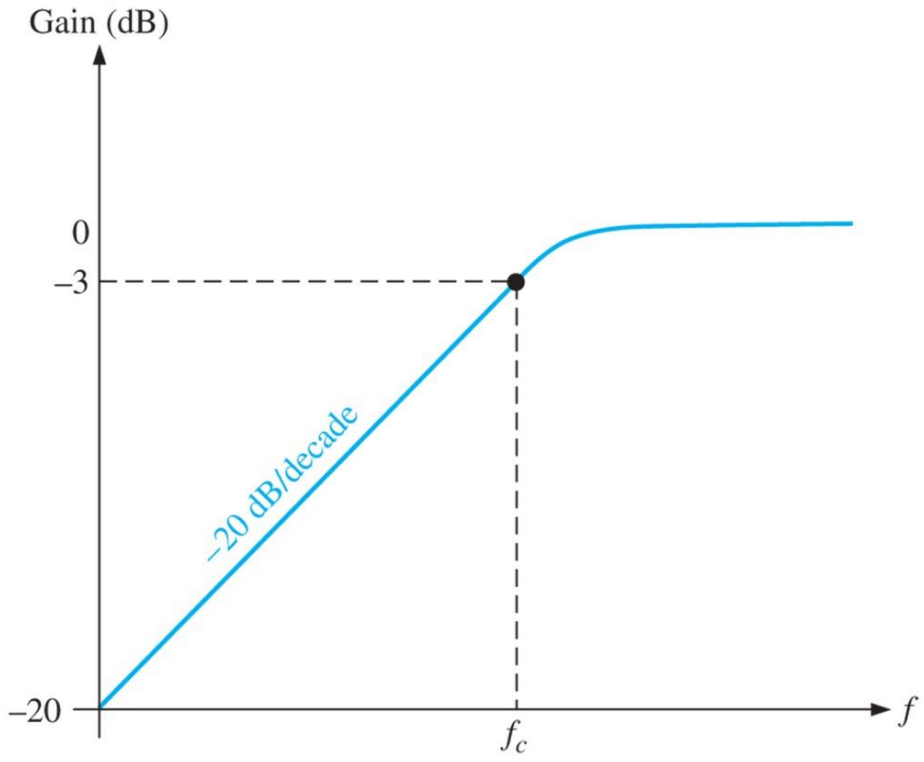
8-4 Active High-Pass Filters

- A single-Pole Filter
- A Two-Pole Filter

A single-Pole Filter



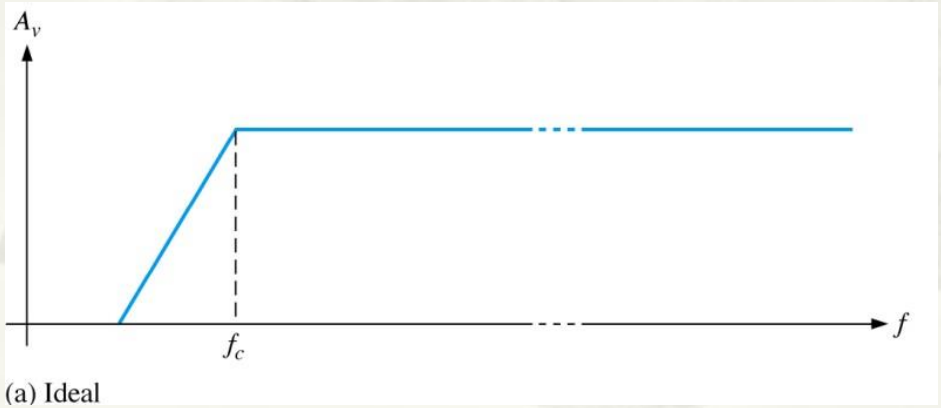
(a)



(b)

A single-Pole Filter

- All op-amps inherently have internal *RC* circuits that limit the amplifier's response at very high frequencies.
- There is an upper-frequency limit on the high-pass filter's response which, in effect, makes it a band-pass filter with a very wide bandwidth.

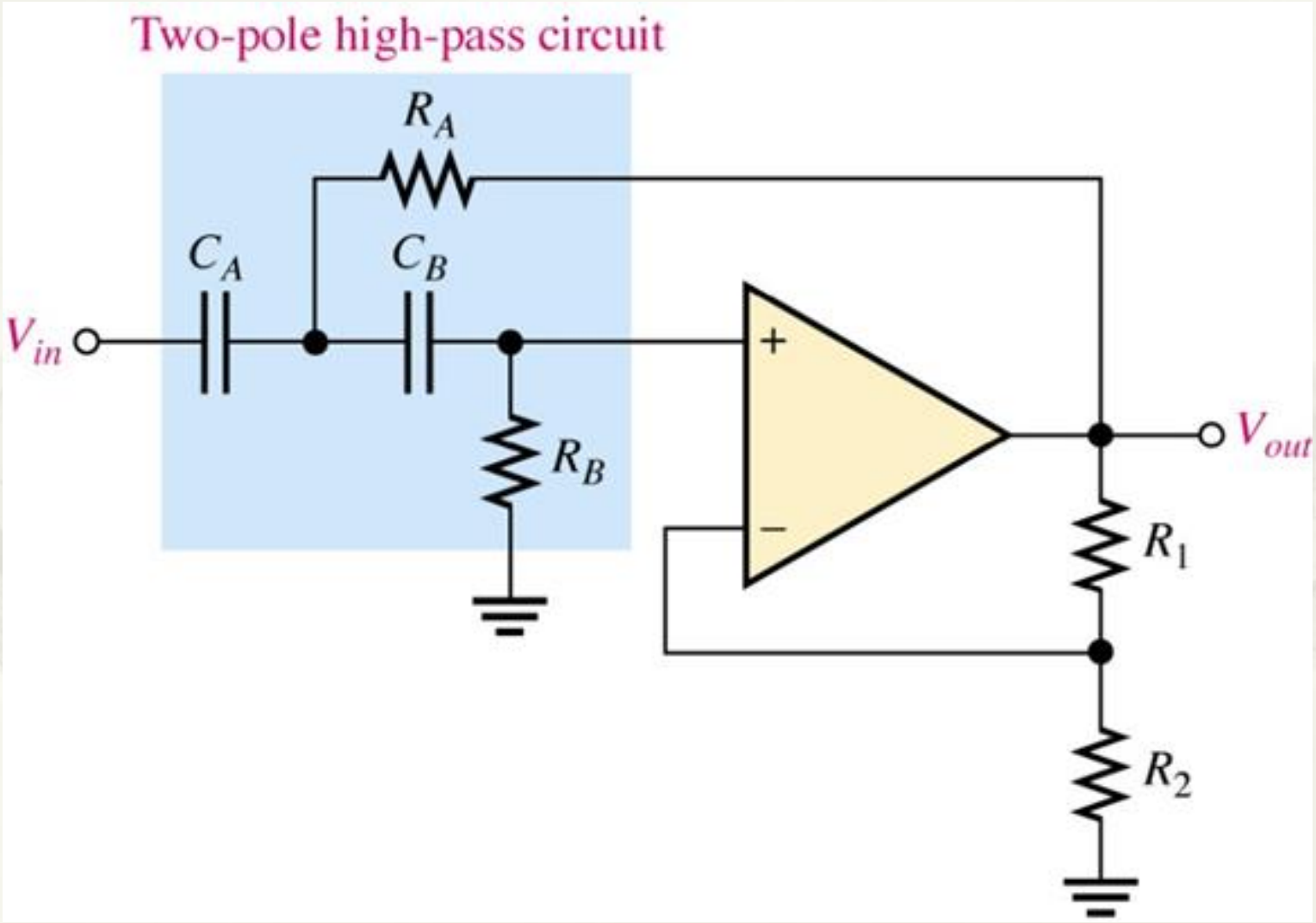


(a) Ideal



(b) Nonideal

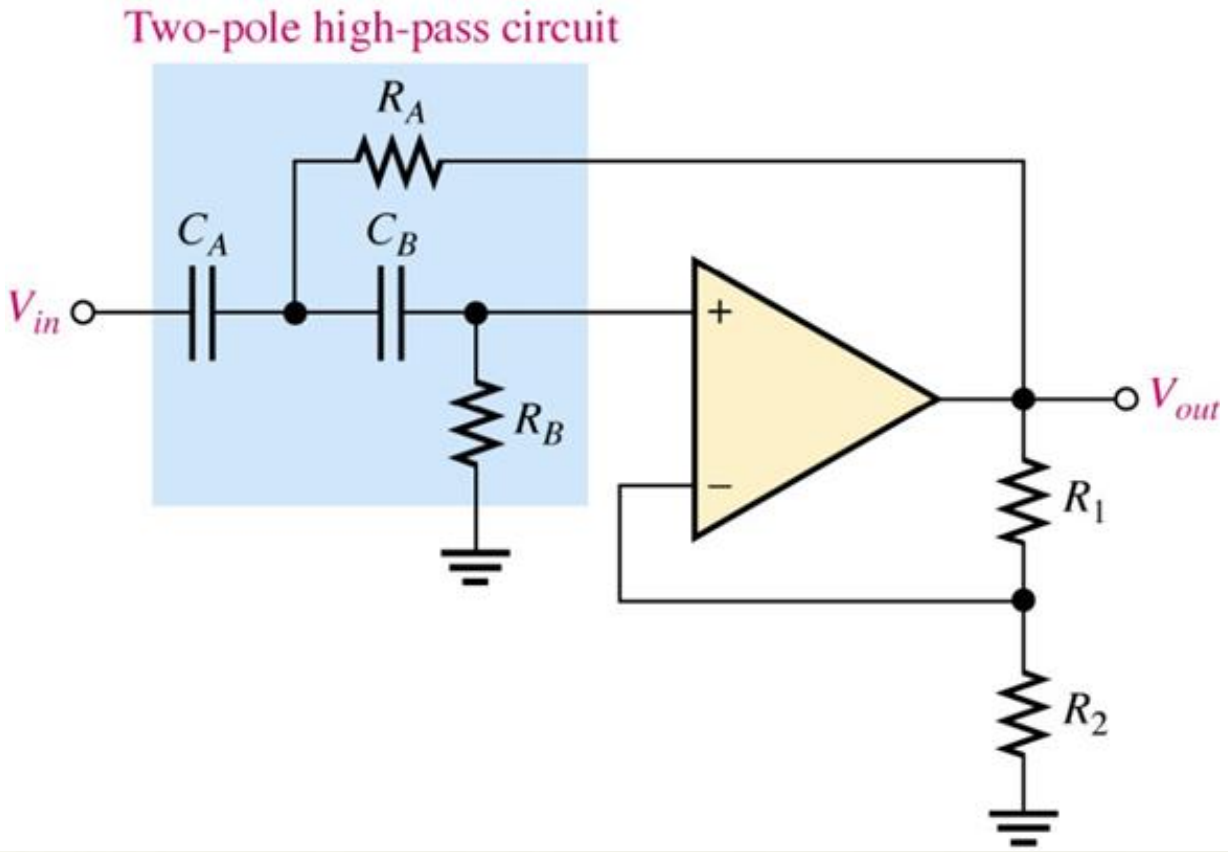
A Two-Pole Filter



Example 8-5

Problem

Choose values for the Sallen-Key high-pass filter in Figure 8–15 to implement an equal-value second-order Butterworth response with a critical frequency of approximately 10 kHz.



Example 8-5

Solution

Start by selecting a value for R_A and R_B (R_1 or R_2 can also be the same value as R_A and R_B for simplicity).

$$R = R_A = R_B = R_2 = \mathbf{3.3\ k\Omega} \quad \text{(an arbitrary selection)}$$

Next, calculate the capacitance value from $f_c = 1/(2\pi RC)$.

$$C = C_A = C_B = \frac{1}{2\pi R f_c} = \frac{1}{2\pi(3.3\ \text{k}\Omega)(10\ \text{kHz})} = \mathbf{0.0048\ \mu F}$$

For a Butterworth response, the damping factor must be 1.414 and $R_1/R_2 = 0.586$.

$$R_1 = 0.586R_2 = 0.586(3.3\ \text{k}\Omega) = \mathbf{1.93\ k\Omega}$$

Example 8-5

Solution

For a Butterworth response, the damping factor must be 1.414 and $R_1/R_2 = 0.586$.

$$R_1 = 0.586R_2 = 0.586(3.3\ \text{k}\Omega) = \mathbf{1.93\ k\Omega}$$

If you had chosen $R_1 = 3.3\ \text{k}\Omega$, then

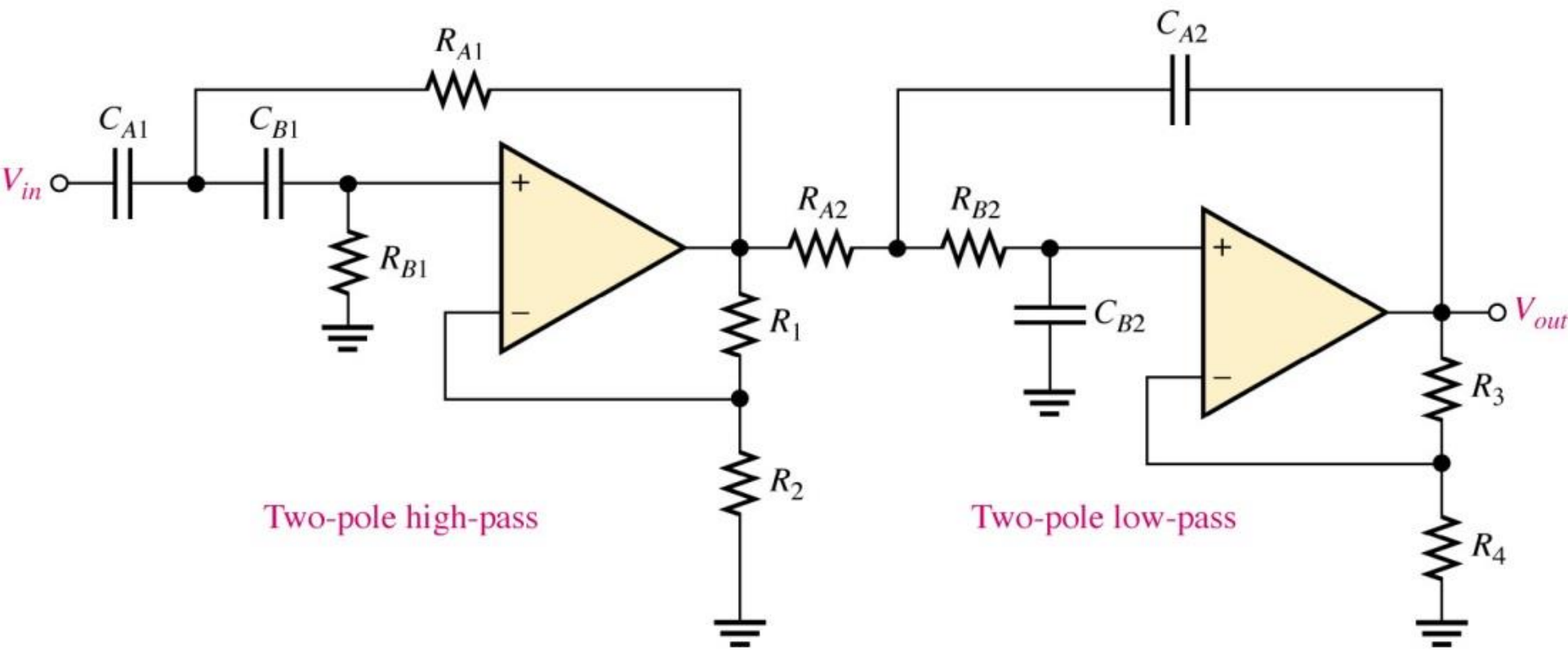
$$R_2 = \frac{R_1}{0.586} = \frac{3.3\ \text{k}\Omega}{0.586} = 5.63\ \text{k}\Omega$$

Either way, an approximate Butterworth response is realized by choosing the nearest standard values.

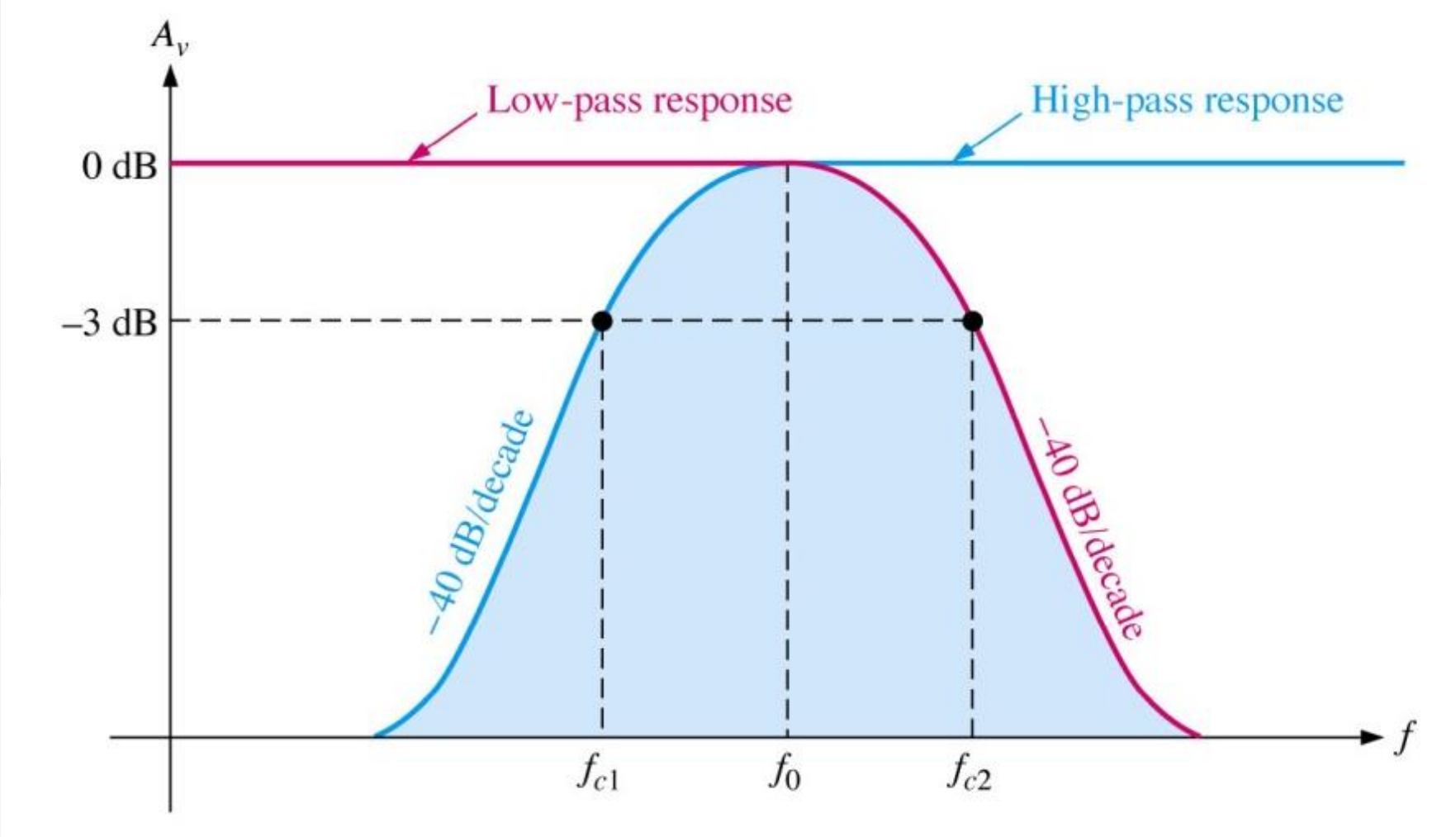
8-5 Active Band-Pass And Band-Stop Filters

- Cascaded Low-Pass and High-Pass Filters Achieve a Band-Pass Response
- Multiple-Feedback Band-Pass Filter
- Multiple-Feedback Band-Stop Filter
- State-Variable Filter

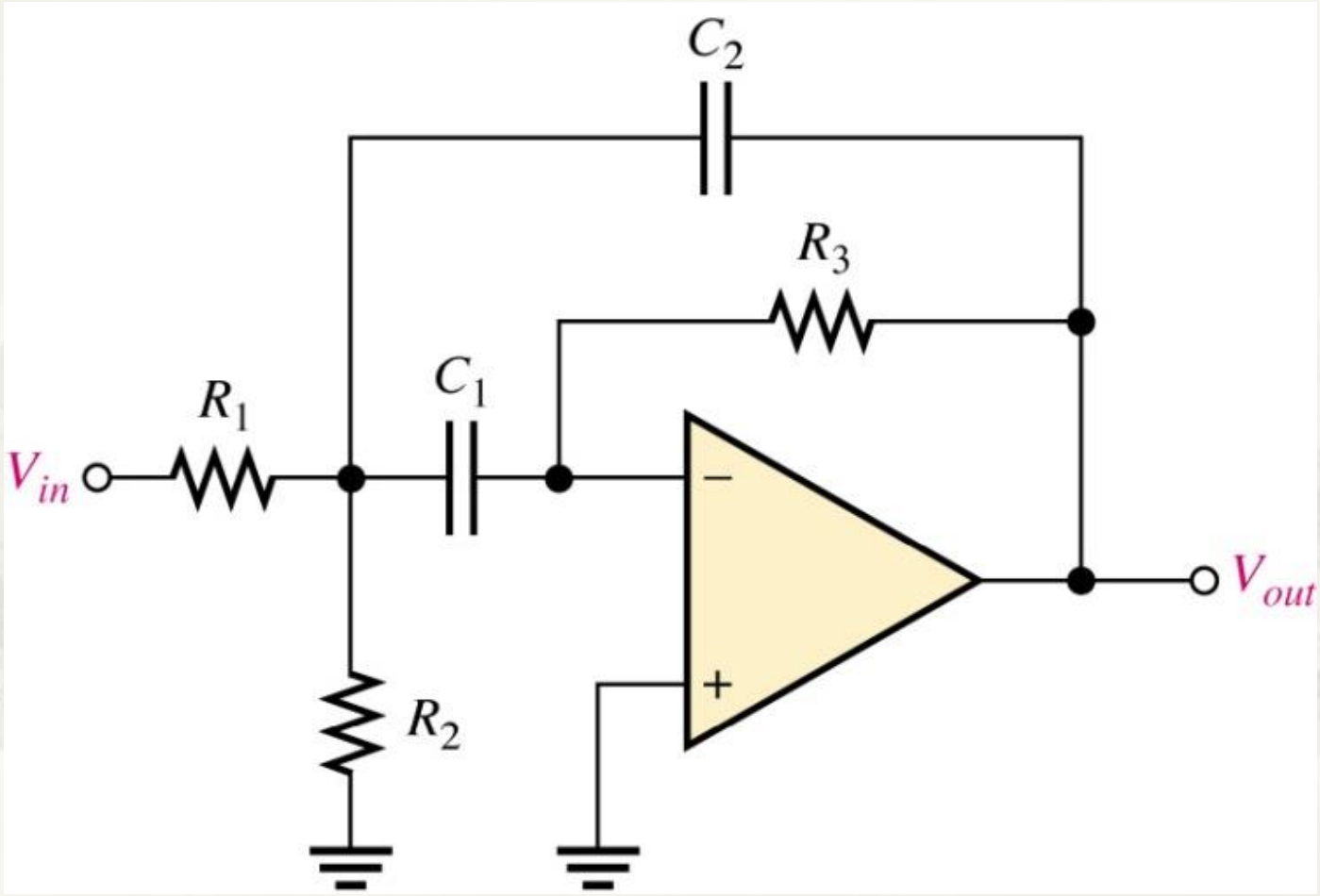
Cascaded Low-Pass and High-Pass Filters Achieve a Band-Pass Response



Cascaded Low-Pass and High-Pass Filters Achieve a Band-Pass Response



Multiple-Feedback Band-Pass Filter



Which is simpler and can achieve narrow bandwidths.

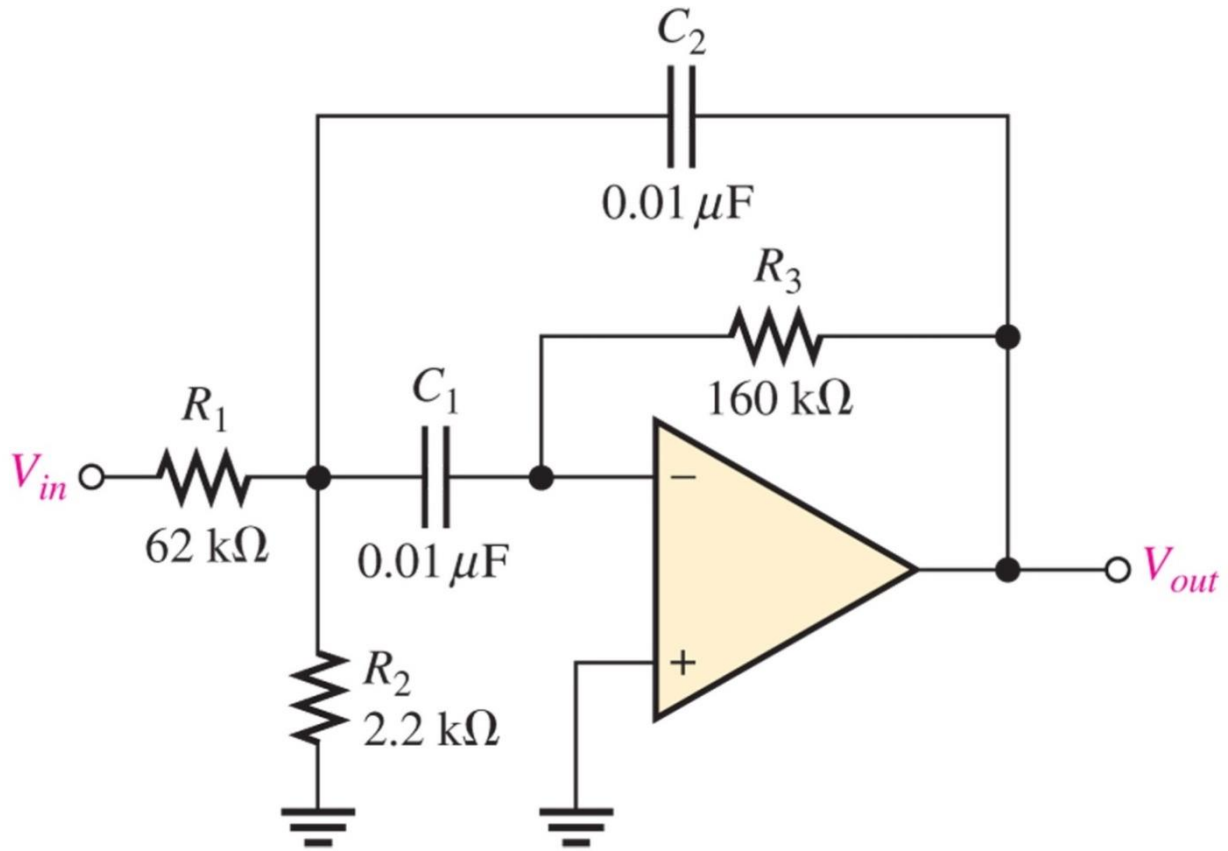
Multiple-Feedback Band-Pass Filter

- **Center frequency**
  - $f_0 = \frac{1}{2\pi C}\sqrt{\frac{R_1+R_2}{R_1 R_2 R_3}}$ where $C = C_1 = C_2$
- **Gain at the center frequency**
  - $A_0 = \frac{R_3}{2R_1}$
- **Bandwidth**
  - $Q = \pi f_0 C R_3$
  - $BW = \frac{f_0}{Q}$

Example 8-6

Problem

Find the center frequency, the gain at the center frequency, and the bandwidth for the filter in Figure 8–18.



Example 8-6

Solution

The center frequency is

$$f_0 = \frac{1}{2\pi C}\sqrt{\frac{R_1 + R_2}{R_1 R_2 R_3}} = \frac{1}{2\pi(0.01\ \mu\text{F})}\sqrt{\frac{62\ \text{k}\Omega + 2.2\ \text{k}\Omega}{(62\ \text{k}\Omega)(2.2\ \text{k}\Omega)(160\ \text{k}\Omega)}} = \mathbf{863\ Hz}$$

The gain is

$$A_0 = \frac{R_3}{2R_1} = \frac{160\ \text{k}\Omega}{2(62\ \text{k}\Omega)} = \mathbf{1.29}$$

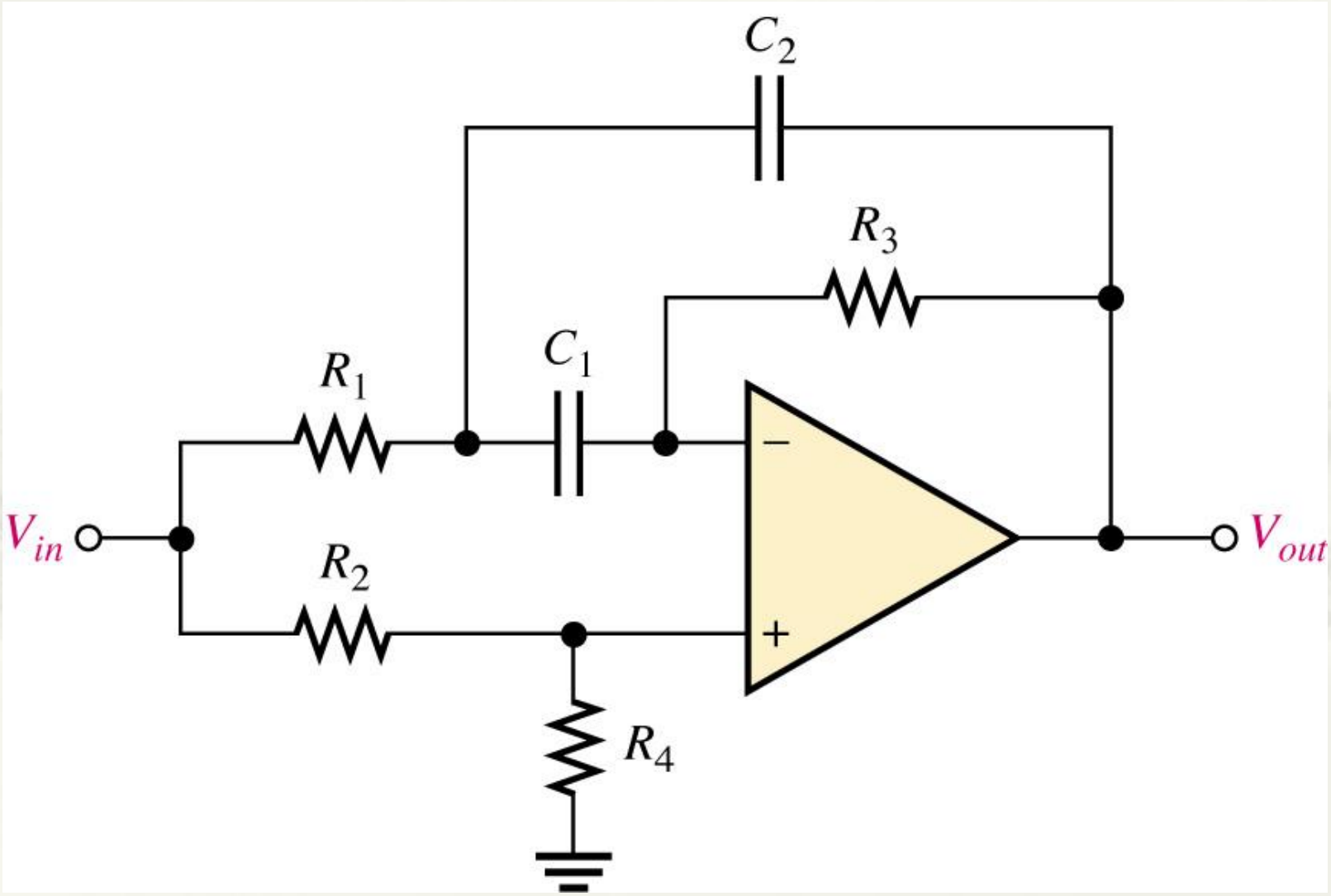
To get the bandwidth, first calculate the Q.

$$Q = \pi f_0 C R_3 = \pi(863\ \text{Hz})(0.01\ \mu\text{F})(160\ \text{k}\Omega) = 4.34$$

Then the bandwidth is

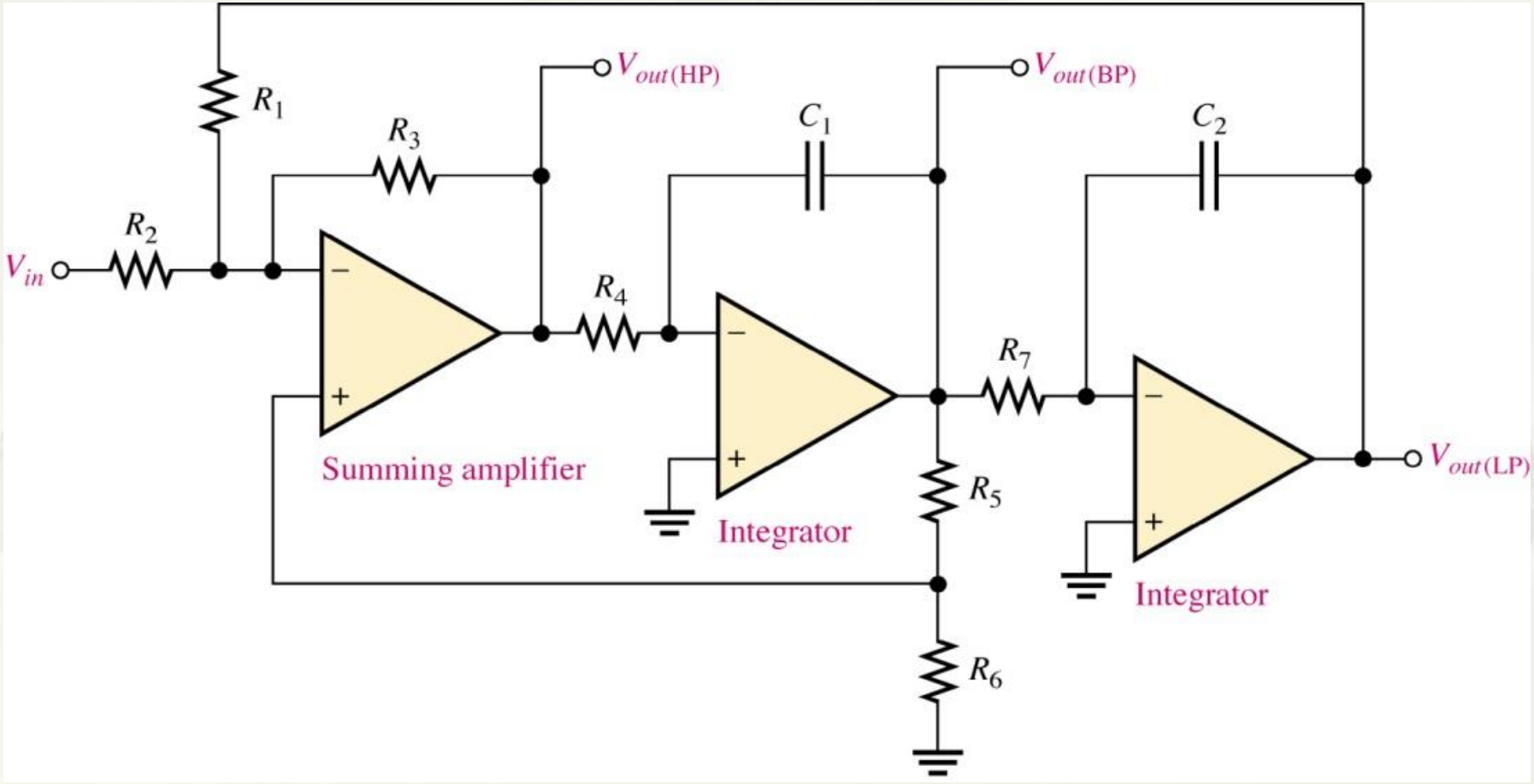
$$BW = \frac{f_0}{Q} = \frac{863\ \text{Hz}}{4.34} = \mathbf{199\ Hz}$$

Multiple-Feedback Band-Stop Filter



State-Variable Filter

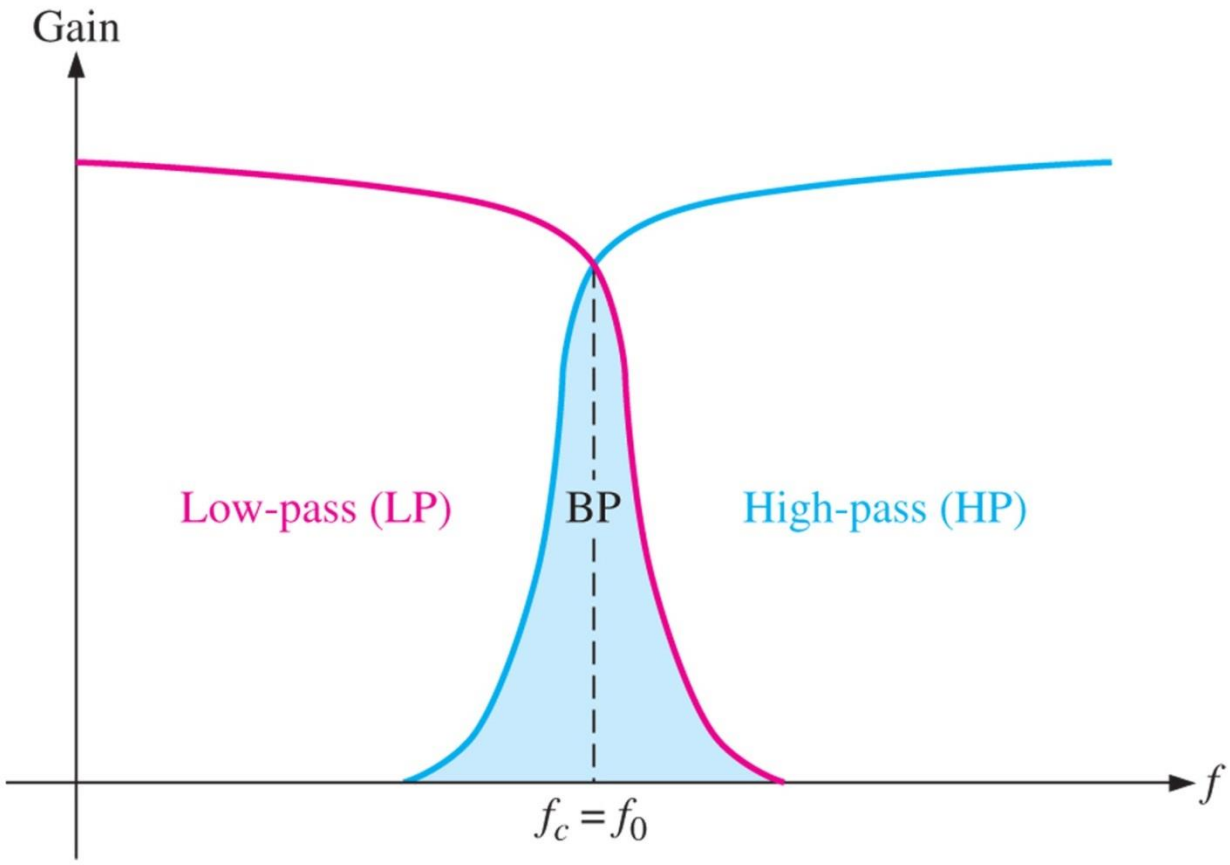
- The state-variable or universal active filter



MF10 datasheet

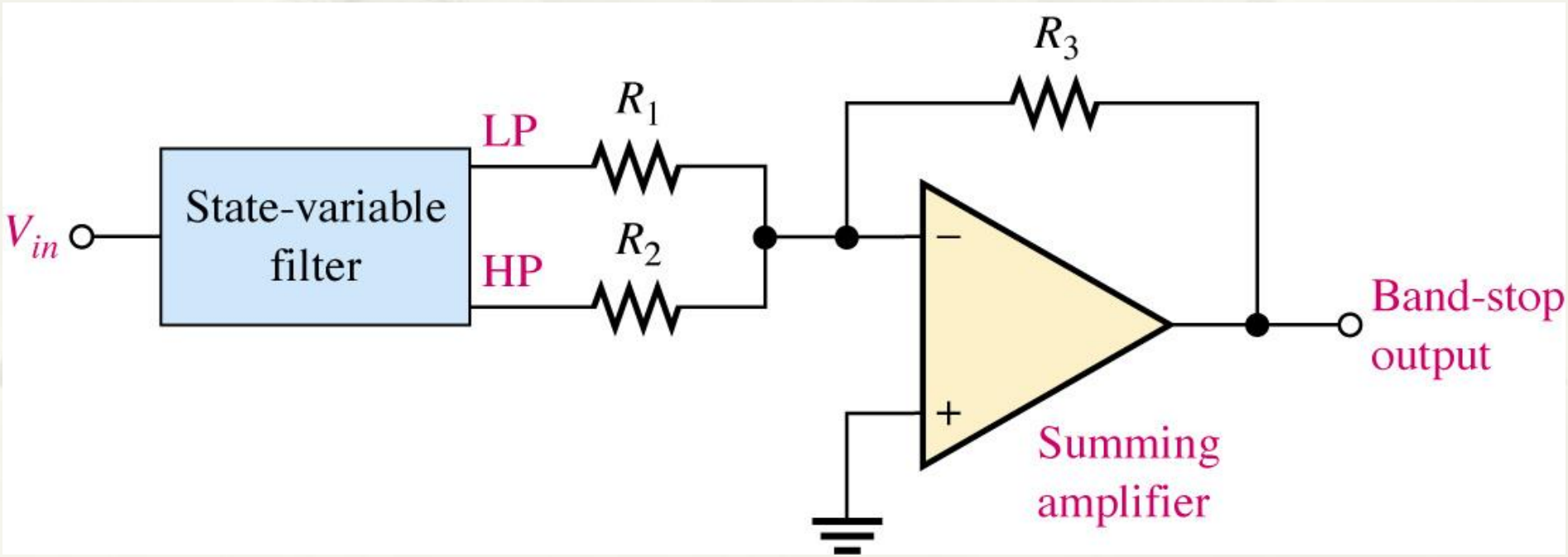
State-Variable Filter

- **Basic Operation**



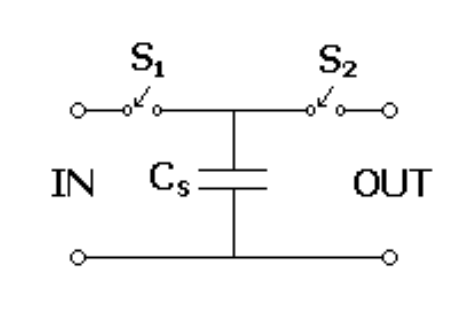
State-Variable Filter

- State-Variable Band-Stop Filter

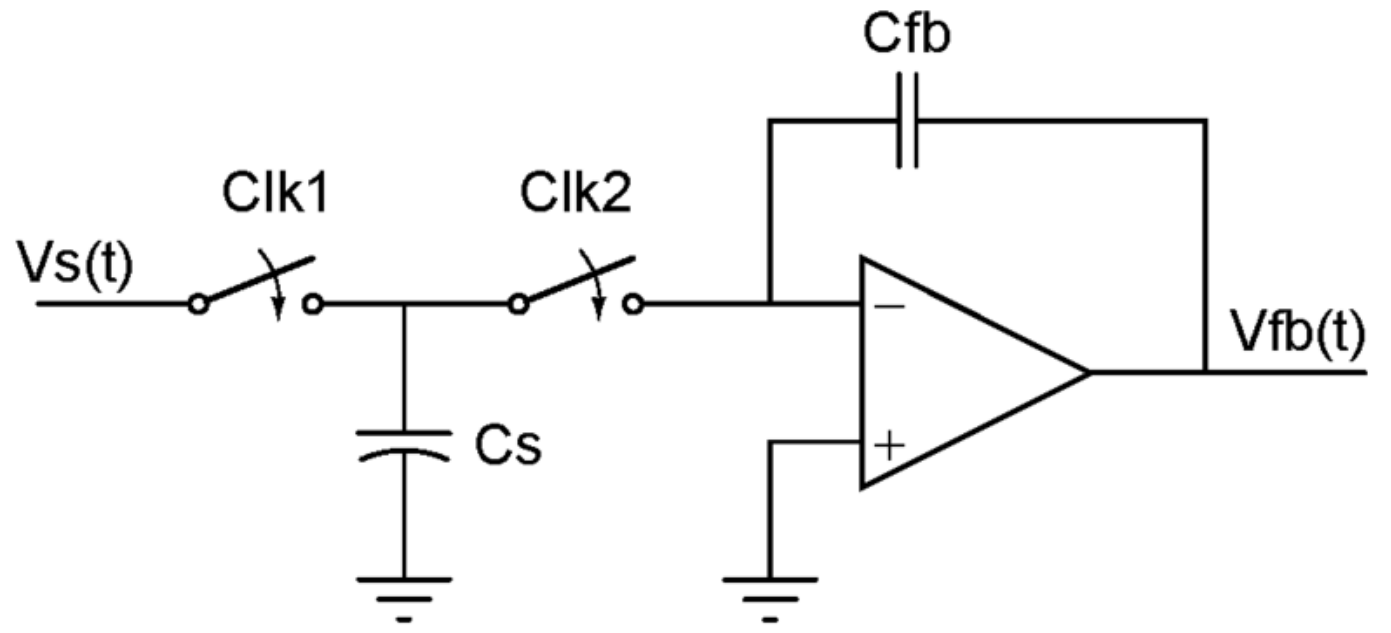


Switched Capacitor Filter

- **Switched capacitor resistor**
  - $R = \frac{1}{C_S f}$



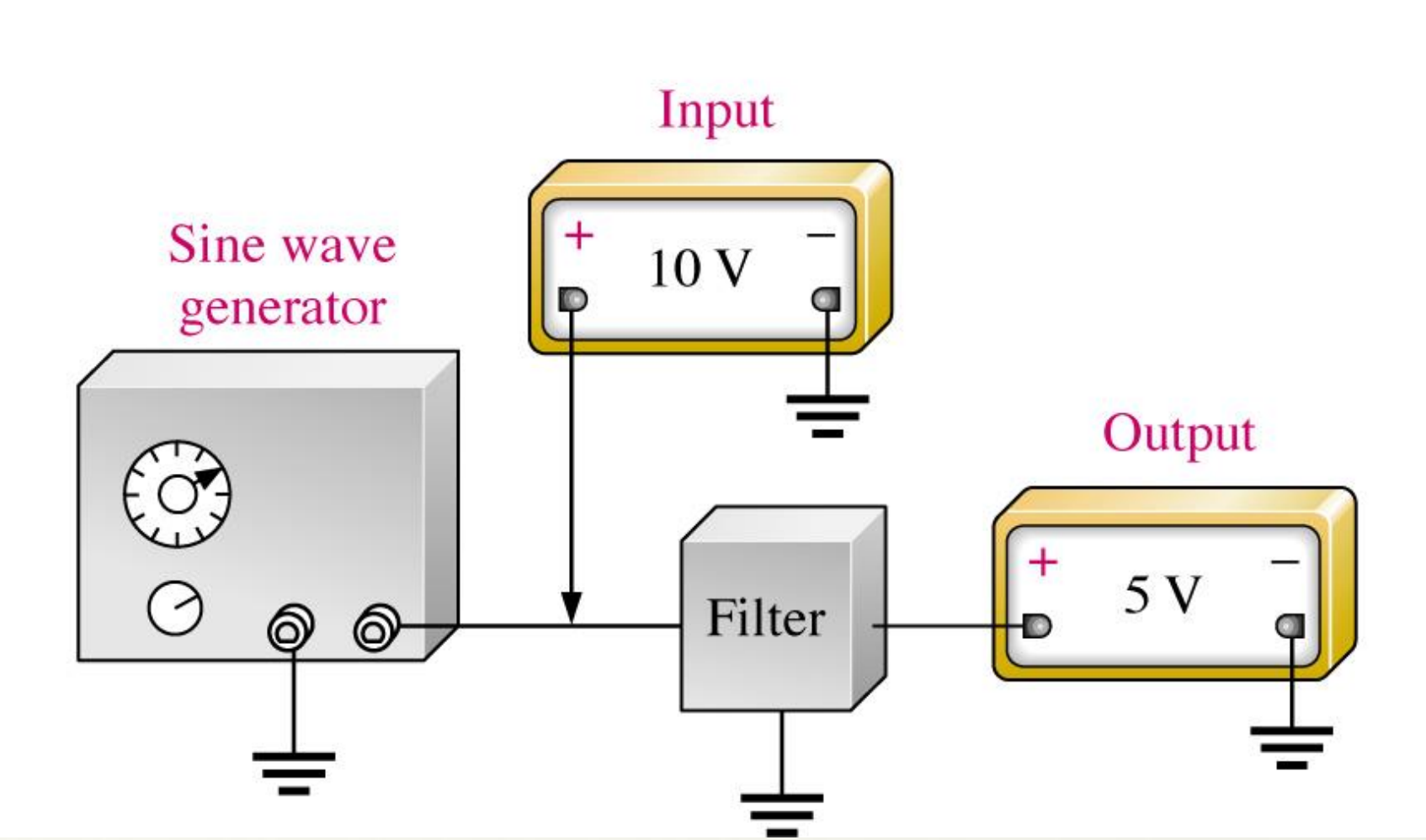
- **Switched capacitor filter (SCF)**



8-6 Filter Response Measurements

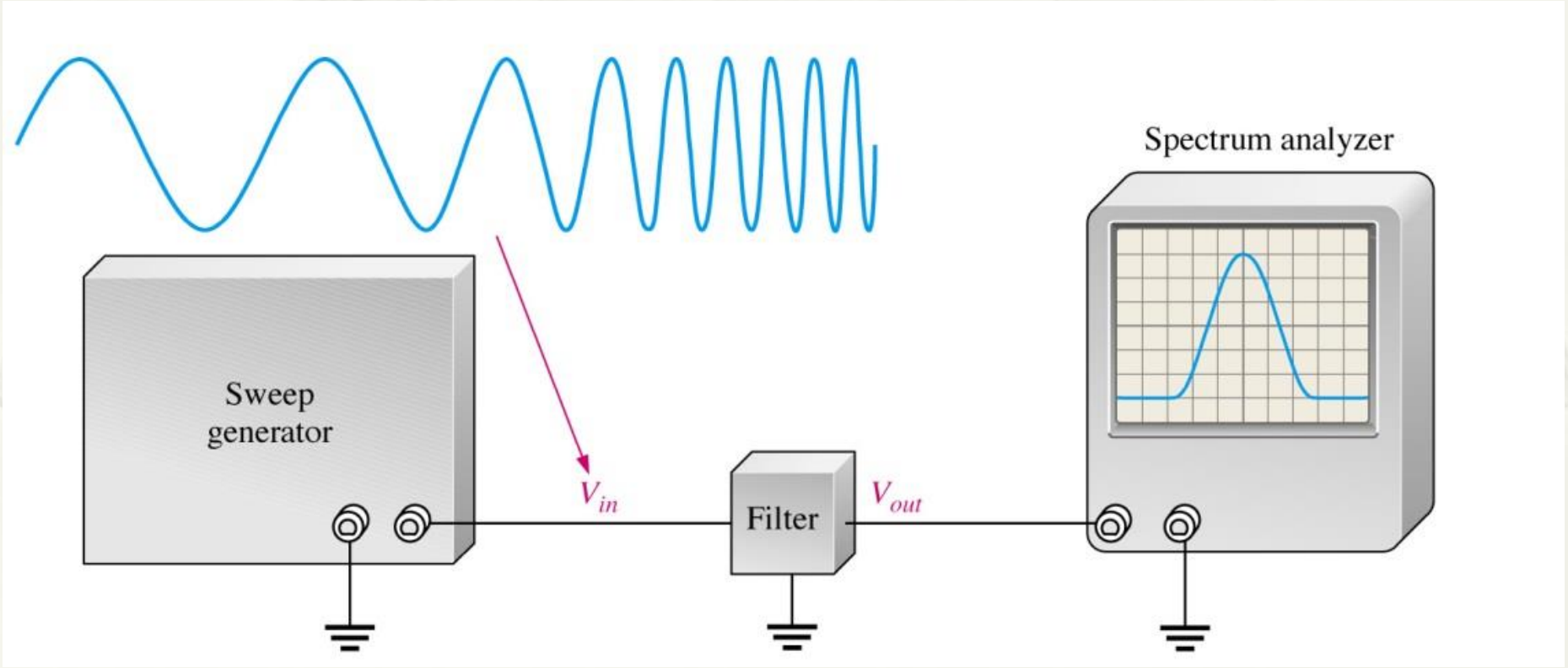
- Discrete Point Measurement
- Swept Frequency Measurement

Discrete Point Measurement



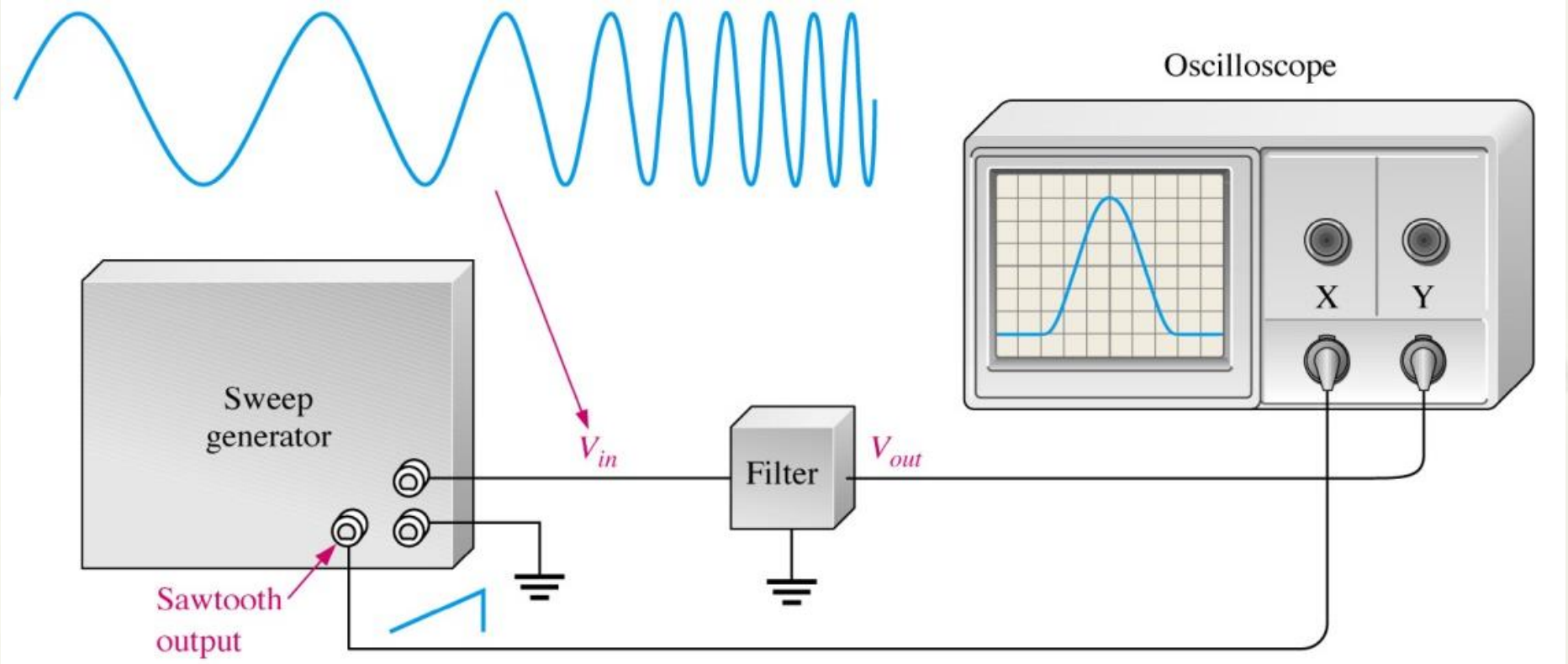
Swept Frequency Measurement

- Use a swept frequency generator and a spectrum analyzer



Swept Frequency Measurement

- Use a swept frequency generator and a oscilloscope



KEY OBJECTIVES

1. Describe the gain-versus-frequency responses of the basic filters
2. Describe the three basic filter response characteristics and other filter parameters
3. Explain how active low-pass filters work
4. Explain how active high-pass filters work
5. Explain how active band-pass and band-stop filters work
6. Discuss two methods for measuring frequency response

HOMEWORK

- **Chapter Checkup (all)**
  - Just do on your textbook
- **Questions (all)**
  - Odd-numbered in your mind, even-numbered on exercise book
- **Basic Problems**
  - 3, 5, 7, 8, 9, 10, 12
- **Basic-plus Problems**
  - 16, 17, 18